ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician
(MAR TECH) (00379)

USQ Engineering Watchkeeper VIC
NQUAL

MITE IDs:

Submarine SCC/MCC Watchkeeper: 122629
OJPR: 122630

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC(A).

Revision	Approval Date	Authority
V 1.0	Jul 01	Provisional Approval DMTE
V 1.1	26 Mar 07	DMTE
V 2.0	25 Mar 19	NTDC(A)
V2.1	###	NPTG

TABLE OF CONTENTS

Title		Page
FOREWORD		v
CHAPTER 1 : TRAINING MANAGEMENT DETAILS		1
	Aim	2
	Course Duration	2
	Resources	2
	Course Capacity	3
	MITE (Loading, Scheduling, Prerequisites, Qualification)	3
	Course Reporting	4
	Evaluation	4
	Validation	4
	Qualification	4
	Prior Learning Assessment and Recognition (PLAR)	4
	Special Instructions	4
CHAPTER 2 : PERFORMANCE OBJECTIVES		5
PO 001	Maintain Submarine Readiness	6
PO 002	Operate Propulsion and Generation Systems	7
PO 003	Operate Main Mechanical Systems	8
PO 004	Operate Electrical Systems	9
CHAPTER 3 : ASSESSMENT OF TRAINEE		10
	General	11
	Enabling and Performance Checks	11
	Responsibility for Testing	11
	Progress Monitoring	12
	Training Review Board	12
	Course Reports	
CHAPTER 4 : LESSON SPECIFICATIONS		13
	Aim	14
	Outline of Training	14
	Training Strategy	14
	Occupation Requirement Organization	14
EO 001.01	Perform Watchkeeper Administrative Duties	15
EC 001.01	Perform Watchkeeper Administrative Duties	18
EO 001.02	Conduct Engineering Rounds	20
EC 001.02	Conduct Engineering Rounds	23
PC 001	Maintain Submarine Readiness	27
EO 002.01	Operate the Diesel Generator	28
EC 002.01	Operate the Diesel Generator	31
EO 002.02	Operate the Main Battery	34
EC 002.02	Operate the Main Battery	36

EO 002.03	Operate the Main Motor	38
EC 002.03	Operate the Main Motor	41
PC 002	Operate Propulsion and Generation Systems	44
EO 003.01	Operate the Cooling Water Systems	47
EC 003.01	Operate the Cooling Water Systems	50
EO 003.02	Operate the Hydraulic Systems	51
EC 003.02	Operate the Hydraulic Systems	54
EO 003.03	Operate the Air Systems	55
EC 003.03	Operate the Air Systems	58
EO 003.04	Operate Trim, Bilge and Ballast Systems	59
EC 003.04	Operate Trim, Bilge and Ballast Systems	62
PC 003	Operate Main Mechanical Systems	64
EO 004.01	Operate Warning Systems	65
EC 004.01	Operate Warning Systems	67
EO 004.02	Operate the Variable Pressure Direct Current (VPDC)	68
EC 004.02	Operate the Variable Pressure Direct Current (VPDC)	70
EO 004.03	Operate the Alternating Current (AC) Systems	72
EC 004.03	Operate the Alternating Current (AC) Systems	74
PC 004	Operate Electrical Systems	76
ANNEX A	**MAIN REFERENCES**	A-1
ANNEX B	**TASK LIST**	B-1
ANNEX C	**QSP WORKING GROUP RECORD OF PROCEEDINGS**	C-1
ANNEX D	**QSP SCALARS**	D-1
ANNEX E	**ON-JOB PERFORMANCE RECORD (OJPR)**	E-1
ANNEX F	**ACRONYMS OR ABBREVIATIONS**	F-1
ANNEX G	**MODULARIZATION AND SUGGESTED TIMINGS**	G-1
ANNEX H	**ADDITIONAL TRAINING RESOURCES**	H-1
ANNEX I	**COSTING INFORMATION**	I-1
ANNEX J	**COURSE HISTORY**	J-1
ANNEX K	**ASSESSMENT PLAN**	K-1
ANNEX L	**TEL DEVELOPMENT**	L-1

FOREWORD

1. This training document is developed under the authority of CNPTG and in accordance with (IAW) the following references:

a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-003/PQ-001 Victoria Class SCC/MCC Watchkeeper Specialty Specification dated 16 February, 2012;

c. A-PD-055-003/PQ-001 Submarine Electrical Technician Specialty Specification dated 14 January, 2009;

d. A-PD-055-003/PQ-001 Submarine Mechanical Maintainer Specialty Specification dated 31 January, 2012;

e. NAVORD 4500-0; and

f. NPTORD 4500-22.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to TDC(A), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training is indicated below:

TE	Module	MITE course ID	NQual
Naval Fleet School Atlantic	Submarine SCC/MCC Watchkeeper	122629	XXXX
Submarine Division	OJPR	122630	XXXX

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Victoria Class Engineering Watchkeeper, Victoria Class Electrical Maintainer, and the Victoria Class Mechanical Maintainer. This QSP is designed to assist NTDC(A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, para 4.5. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	241
Enabling Checks (ECs)	33
Performance Checks (PCs)	14
Actual Training Time (ATT) Sub-total	288 49 **trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	11
Training Administrative Time (TAT)	11
Training Travel Time (TTT)	0
Physical Fitness (PT)	27
Sub-total	39 7 **trg days**
Course Duration Calculation (<u>ATT+BAT+TTT+TAT+PT</u>) **7 (50 min) periods** * **Partial days must be rounded up**	337 49 **trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional TE resources can be listed in Annex H, if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO/EO
Course Officer	Lt (N)	MSEO	1	
Course Senior Instructor	PO1	MARTECH – RQ PO1/ALUV	1	
Course Instructor	MS-PO2	MARTECH – USQ S1 MECH/ALUS	1	
Course Instructor	MS-PO2	MAR TECH – USQ S1 ELEC/ALUT	1	

Critical Training Resources	**No of Days**
Classroom B9 (S-15) & Imperial Classroom (S-100)	43
MCT Trainer (S-100)*	23
SCT Trainer (S-100)*	16

Submarine Diesel Trainer (S15)	1
Electrical Trainer Room (S-100)	1

* Can be run concurrently.

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

a. Maximum – 8;

b. Minimum – 1; and

c. Optimum – 6.

MILITARY INDIVIDUAL TRAINING AND EDUCATION (MITE)

1. Loading. The authority to load members onto this course lies with the TE. NPTG is to be included as an INFO addressee on all loading messages.

2. Scheduling. Each session of this course shall be scheduled, in MITE, IAW NPTORD 4500-1 and NPTORD 4500-40 by the TE conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

3. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

d. AILS qualified;

e. MAR TECH USQ S1 qualified;

f. Security clearance level II; and

g. Medically fit Submarines.

4. Qualification. Successful completion of the modules constitutes the primary means of attaining the qualifications as listed. The qualification is awarded upon completion of the associated OJPR.

h. USQ Engineering Watchkeeper VIC (NQUAL):

(1) Module 1, Residential: 122629; and

(2) Module 2, OJPR: 122630.

COURSE REPORTING

5. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW DAOD 3031-9 and NPTORD 4500-51.

EVALUATION

6. Program evaluation shall be conducted IAW the guidance provided in CFITES Vol. 11.

VALIDATION

7. This training will be validated every five years CFITES Vol. 8 and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

8. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

9. This QSP is divided into Modules. Module 1 is the Submarine SCC/MCC Watchkeeper course and Module 2 is the OJPR to be completed upon completion on Module 1.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (C/NPTG)

PO 001 Maintain Submarine Readiness

1. **Performance**: Maintain submarine readiness.

2. **Conditions**:

   a. Given:

      (1) Submarine;
      (2) Access to DRMIS;
      (3) Access to Serv+;
      (4) Lockout/Tagout devices and materiel;
      (5) Engineering logs;
      (6) References; and
      (7) Assistance/Supervision as required.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) All weather conditions;
      (3) Surfaced or submerged; and
      (4) All light conditions.

3. **Standard**: The SCC/MCC Watchkeeper shall maintain the submarine readiness, IAW refs, by:

   a. Performing watchkeeper administrative duties, IAW refs, to include:

      (1) Conducting lockout/tagout IAW refs A5, A7 and A8;
      (2) Complying to environment policies IAW ref A6; and
      (3) Preparing preventive/corrective maintenance documentation (AFT0400), IAW C5, to include complying to first level system requirement IAW ref A7.

   b. Conducting Engineering Rounds, IAW refs A3, A4 and A5, to include:

      (1) Conducting Pre-sail checks (ZT0020);
      (2) Performing diving checks;
      (3) Conducting Pre-watch rounds, to include conducting escape rounds, IAW refs; and
      (4) Conducting on-watch rounds, to include maintaining engineering logs (AKT1090).

4. **Training Limitations**: Nil.

PO 002 Operate Propulsion and Generation Systems

1. **Performance**: Operate propulsion and generation systems.

2. **Conditions**:

   a. Given:

      (1) Operational Submarine;
      (2) Operational Diesel Engines;
      (3) Operational Main Battery;
      (4) Operational Propulsion Switchboard or Propulsion Switchboard Trainer;
      (5) References; and
      (6) Assistance/Supervision as required.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) All weather conditions; and
      (3) Surfaced or submerged.

3. **Standard**: The SCC/MCC Watchkeeper shall operate the propulsion and generation systems, by:

   a. Operating the diesel generator, to include:

      (1) Preparing the diesel-electric systems for starting/stopping, IAW refs A5 and B7;
      (2) Starting/Stopping and monitoring correct local and remote operation of diesel-electric systems (QT0390, QT0350, QT0360, QT0320, QT0310, QT0320, PT0280, PT0290), IAW refs A5 and B7; and
      (3) Conducting Emergency Operating Procedures (EOP's) (YT0200), IAW ref A1.

   b. Operating the main battery, IAW refs A5, B4, and B25 to include:

      (1) Maintaining state boards, IAW refs; and
      (2) Maintaining battery log (AUT4090), IAW refs.

   c. Operating the main motor (PT0310, IT 0470), IAW refs A5 and B8, to include operating the shaft line (YT0160, YT0170), IAW refs A5, B9, B13 and C1.

4. **Training Limitations**: Nil.

PO 003 Operate Main Mechanical Systems

1. **Performance**: Operate main mechanical systems

2. **Conditions**:

   a. Given:

      (1) Operational Submarine;
      (2) Systems in correct SOP lineup;
      (3) References; and
      (4) Assistance/Supervision as required.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) All weather conditions; and
      (3) Surfaced or submerged.

3. **Standard**: The SCC/MCC Watchkeeper shall operate main mechanical systems, by:

   a. Operating the cooling water systems (DT0200, JT0260, YT0090), to include:

      (1) Operating the sea water cooling system, IAW refs A5 and B15;
      (2) Operating the fresh water cooling system, IAW refs A5 and B15; and
      (3) Operating the chilled water cooling system, IAW refs A5 and B12, to include operating ventilation systems (YT0180,YT0190), IAW A5 and B11.

   b. Operating the hydraulic systems (YT0110), to include:

      (1) Operating the main, external, and steering and hydroplanes plants IAW refs A5 and B13;
      (2) Operating the main and external hydraulic systems IAW refs A5 and B13;
      (3) Operating steering and hydroplanes IAW refs A5 and B13; and
      (4) Performing a Bathythermograph, IAW refs A10.

   c. Operating air systems (YT0120, YT0140), IAW refs A5 and B10, to include carrying out vacuum test, IAW refs A5; and

   d. Operating trim, bilge and ballast systems (YT0070, YT0050), IAW refs A5 and B14 to include:

      (1) Operating the HP bilge and ballast systems;
      (2) Operating the trim system; and
      (3) Operating the LP bilge system.

4. **Training Limitations**: Nil.

PO 004 Operate Electrical Systems

1. **Performance**: Operate electrical systems.

2. **Conditions**:

   a. Given:

      (1) Operational Submarine;
      (2) References; and
      (3) Assistance/Supervision as required.

   b. Denied: Nil.

   c. Environment:

      (1) At sea or alongside;
      (2) All weather conditions; and
      (3) Surfaced or submerged.

3. **Standard**: The SCC/MCC Watchkeeper shall operate the electrical systems IAW refs, by:

   a. Operate the warning systems (YT0210, YT0220) to include:

      (1) 24V Grade A and B, and Flood alarms, IAW refs A5 and B5;
      (2) Fire Detection System, IAW refs C6; and
      (3) Central Surveillance System, IAW C5.

   b. Operate the VPDC (AUT1250, PT0140, PT0300), IAW refs A5 and B3, to include:

      (1) Power breakers;
      (2) Auxiliary DC equipment; and
      (3) Battery switchboard.

   c. Operating the Alternating Current (AC) distribution system (PT0230, PT0240, PT0120), IAW refs A5 and B6 to include operating the 24V DC distribution system (PT0110) IAW A5 and B5.

4. **Training Limitations**: Nil.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDING OFFICER NAVAL TRAINING DEVELOPMENT CENTER ATLANTIC (CO NTDC(A))

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this Qualification Standard and Plan (QSP) and the standard listed must be met to achieve this qualification. PCs will be administered in the form of a performance-based assessment and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

10. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructors and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

11. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

12. Assessment of trainees is to be done in accordance with (IAW) CFITES Vol. 7.

13. The Training Establishments (TE) provide quality control by designing, administering and scoring all Performance Checks. Any PC not administered or invigilated by standards staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

14. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol. 6, and TE SOs or SOPs.

15. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the Training Review Board (TRB) or Progress Review Board (PRB) (if required); and

d. Information to assist in completion of the CF 377 Course Report.

16. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs.

17. PC failures and supplemental PC failures are to be handled IAW TE SOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

18. TRBs shall be convened and conducted IAW with NPTORD 4500-54.

COURSE REPORTS

19. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 and NPTORD 4500-51.

20. Trainee will be graded pass or fail. Students will not be ranked IAW NPTORD 4500-51.

21. Where necessary for Rank Qualifications (RQ), the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the TE.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDING OFFICER NAVAL TRAINING DEVELOPMENT CENTER ATLANTIC (CO NTDC(A))

AIM

1. The aim of the training resulting from this Training Plan (TP) is to prepare personnel who can perform the duties of an USQ Engineering Watchkeeper VIC.

OUTLINE OF TRAINING

2. This TP contains Lesson Specifications with associated Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) that the candidate shall achieve.

TRAINING STRATEGY

3. The training requirement will be achieved through the following modules:

Module	MITE ID	Reg/P Res	Days[1]	Min	Max	NQuals
Mod 1: SCC/MCC Watchkeeper	122629	Reg	42	1	6	XXXX
Mod 2: OJPR	122630	Reg	180			XXXX

Note 1. Only includes EO + EC + PC time.

4. Further details can be found at Annex G – Modularization and Suggested Timings.

PO 001 Maintain Submarine Readiness

EO 001.01 Perform Watchkeeper Administrative Duties

1. **Performance**: Perform watchkeeper administrative duties

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Access to DRMIS and Serv+;
      (4) Lockout/Tagout devices;
      (5) Lesson Plan; and
      (6) References.

   b. Denied: Nil;

   c. Environment: Classroom.

3. **Standard**: The trainee shall perform the watchkeeper administrative duties, IAW refs, to include:

   a. Conducting lockout/tagout IAW refs A5 and A6;

   b. Complying to environment policies IAW refs A6; and

   c. Preparing preventive/corrective maintenance documentation, IAW refs C5, to include complying to first level system requirement A7.

4. **Time Required**:

   a. EO – 6 x 50 min periods; and

   b. EC – 3 x 50 min periods.

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Self-study (Reading Assignment); and

   c. EDI.

6. **Training Aids**:

   a. SSOs;

   b. Handout forms; and

   c. Lockout devises.

7. **Test Details:** The EC will be conducted as a tabletop practical exercise. Trainee must attain a minimum mark of 7/10 SATs **AND** all critical elements on the EC to pass the EC. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Conducting lockout/tagout						
1) Conducting the lockout/tagout procedure, to include a) responsibility of personnel, to include: i) Ship's staff; and ii) OOD/DWS. b) Electrical lockouts, c) Mechanical lockouts, and d) Lockout devices.	**25**	**25**	**100**	A5, A6:SOP S4	Interactive Lecture Training aids: - SSOs - Handout forms - Lockout devises	
2) Conducting mast control and man aloft.	**15**	**10**		A6:SOP S4		
Complying to environment policies						
3) Complying to Victoria class environmental policy, in regards to: a) Environmental protection: i) MARPOL special areas; b) Waste/Oily water disposal; c) High risk evolutions including: i) Fuelling, ii) Transferring POL's, iii) HAZMAT, d) Due Diligence.	**25**			A6:SOP E2 – E4	Self-study: Reading Assignment	
Prepare preventive/corrective maintenance documentation, to include comply to first level system requirement.						

4) Describing quality assurance for first level systems, in regards to: a) Purpose, b) Submarine safety, i) Uncontrollable flooding ii) Size of the leak iii) Loss of control, iv) Loss of main propulsion, c) First level definition, d) First level guidelines.	**25**			A7	Self-study: Reading Assignment	
5) Preparing preventative/corrective maintenance documentation, to include: a) Cartidge cards, b) Joint cards, c) Hose cards, d) Objective Quality Evidence (OQE), and e) QCA documentation.	**10**	**15**	**50**	A7	EDI	

Enabling Check – EO 001.01 Perform Watchkeeper Administrative Duties

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. This assessment will be conducted as a tabletop practical exercise in the classroom.

2. The trainees will require access to system drawings, sufficient Lockout/Tagout forms, and sufficient Maintenance Documentation forms. Part of this assessment will be a take home, open book write up detailing Environmental policies with regards to waste and oily water disposal.

Arrangements and Procedures

3. EC 001.01 will be a tabletop practical and take home write up assessment based on content from EO 001.01.

4. To pass this EC, the trainee must achieve 7/10 SATs including all critical SATs.

5. Time required: 3 x 50 min period(s)

6. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

7. EC 001.01 will be a practical assessment based on content from EO 001.01.

8. During this practical assessment, you will be required to Perform Watchkeeper Administrative Duties. You must pass this EC to pass EO 001.01.

9. You will be given a scenario of a repair that needs to be completed. It will have to be physically and electrically locked out to ensure safety while working on the system.

10. References will be provided.

11. Pass mark is 7/10 SATs including all critical SATs IAW ECL 001.01.

12. You will have 3 x 50 min period(s) to complete this assessment.

13. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The assessment is based on the requirement to ensure safety and environmental stewardship during repairs and maintenance.

PART IV – ENABLING CHECK LIST

Item	EC 001.01 Performance Statement	SAT	UNSAT	Remarks
1.	**Conducting Lockout/Tagout**			
a.	Design a lockout to allow for repair of the defect in the scenario. Include the following:			
i.	Safety; all pressure/electric supply is isolated.	☐	☐	**Critical**
ii.	System will not be able to be operated.	☐	☐	**Critical**
iii.	Form is properly filled out.	☐	☐	
iv.	Write a brief to the duty watch, informing them of losses/mitigations required for the Lockout.	☐	☐	
2.	**Maintenance Documentation**			
a.	Prepare the maintenance documentation correctly to include:			
i.	Cartridge card, if required;	☐	☐	
ii.	Joint card, if required;	☐	☐	
iii.	Flex hose card, if required.	☐	☐	
3.	**Environmental Policies**			
a.	State the environmental requirements for the following:			
i.	Disposing of solid garbage at sea;	☐	☐	
ii.	Disposing of HAZMAT at sea and alongside;	☐	☐	
iii.	Disposing of oily water at sea and alongside.	☐	☐	

Score: /

The trainee must achieve 7/10 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 001.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 001.01 with the candidate.		**Candidate** I have been debriefed on the result of EC 001.01.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 001 Maintain Submarine Readiness

EO 001.02 Conduct Engineering Rounds

1. **Performance**: Conduct Engineering Rounds

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom; and
      (2) SCT/MCT.

3. **Standard**: The trainee shall conduct engineering rounds, IAW refs A3, A4, and A5, to include:

   a. Conducting Pre-sail checks, IAW refs;

   b. Performing diving checks, IAW refs;

   c. Conducting Pre-watch rounds, IAW refs, to include conducting escape rounds;

   d. Conducting on-watch rounds, IAW refs to include maintaining engineering logs; and

   e. Effectively communicating cause and effect of equipment and/or system failure/defects.

4. **Time Required**:

   a. EO – 7 x 50 min periods;

   b. EC – 6 x 50 min periods; and

   c. PC – OJPR

5. **Instructional Methods**:

   a. Interactive lecture;

   b. EDI; and

   c. OJPR.

6. **Training Aids**:

   a. Pre-sea checks slide show;

b. MCT/SCT Trainers; and

c. Watch Rounds educational videos.

7. **Test Details**: This EO will be tested as part of EC 001.02 as a practical test, this will concurrently test EO's 001.02, 003.01 – 003.03, and 004.01 and will occur during a tour of an available submarine. In the event that a submarine is unavailable, the assessment will be written. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Conduct Pre-Sail Checks						
1) Conducting pre-sea checks, to include: a) Long; and b) Short.	**25**			A5: Vol. 2, pg. 259 – 288	Interactive Lecture: Training aids: Slide show	
Perform diving checks						
2) Perform diving checks, to include: a) Open Up for Dive; and b) Post Diving Checks, including: i) Long; ii) Short.	**25**	**25**		A5: Vol. 2, pg 241 – 249 and 331 – 335	Interactive Lecture: Scenario-based Training aids: Panel (MCT SCT Trainers)	1 scenario for the class. Max interaction with trainees: it's their scenario.
Conduct Engineering Rounds, to include Pre-watch rounds, escape rounds, and on-watch rounds						
3) Conducting pre-watch rounds, to include: a) Engineering spaces, b) Engineering systems, c) Engineering equipment,	**50**	**25**	**50**	A3, A5	Field Trip: Submarine S1: Educational video.	If submarine unavailable use an educational video to provide the information in an fast and standardize way.

4) Escape rounds, to include: a) Forward escape compartment, b) After escape compartment.	**25**	**10**		A3, A5		
5) Conducting on-watch rounds, to include: a) Engine rounds, b) Safety rounds, c) Auxiliary equipment and spaces.	**25**	**15**		A3, A5		
Effectively communicating cause and effect of equipment and/or system failure/defects Maintain engineering logs						
6) Completing required engineering logs:	**10**			A3, A4	Field Trip	
7) Communicating failures/defects, to include: a) Discovery, b) Logging, c) Reporting,	**15**			A3, A4		

Enabling Check – EO 001.02 Conduct Engineering Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine, field trip.

Arrangements and Procedures

2. EC 001.02 will be a practical assessment based on content from EO 001.02. It will be conducted during a field trip to the submarine concurrently with ECs 003.01 – 003.03, and 004.01. The student will be required to demonstrate their abilities to conduct pre-watch and on watch rounds to the instructor.

3. To pass this EC, the trainee must achieve the following:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

4. Time required: 1 x 50 min period(s) per trainee (6 total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. This walkthrough will be a practical assessment based on content from EO 001.02, 003.01 – 003.03, and 004.01.

7. During this practical assessment, you will be required to conduct Engineering Rounds. You must pass this EC to pass EO 001.02, 003.01 – 003.03, and 004.01.

8. References will not be provided.

9. To pass this EC, the trainee must achieve the following, IAW ECL 001.02:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

Item	EC 001.02, 003.01 – 003.03 AND 004.01 Conduct Engineering Rounds	SAT	UNSAT	Remarks
1.	**Forward of 35 Bulkhead**			
a.	Bilges and Space condition	☐	☐	
b.	Air Systems	☐	☐	
c.	Hydraulic Systems	☐	☐	
d.	Fridges	☐	☐	
e.	Fire Fighting Equipment	☐	☐	
2.	**Between 35 and 56 Bulkhead**			
a.	Bilges and Space condition	☐	☐	
b.	Air Systems	☐	☐	
c.	Hydraulic Systems	☐	☐	
d.	24V DC Systems	☐	☐	
e.	ROD	☐	☐	
f.	Fire Fighting Equipment	☐	☐	
3.	**Aft of 56 Bulkhead**			
a.	Bilges and Space condition	☐	☐	
b.	Air Systems	☐	☐	
c.	Hydraulic Systems	☐	☐	
d.	Electrical systems	☐	☐	
e.	Warning Systems	☐	☐	
f.	Fire Fighting Equipment	☐	☐	
Score: /				
	The trainee must achieve 12/17 SATs to pass EC 001.02			
4.	**Operate the Sea Water and Fresh Water System**			
a.	Properly line up the sea water hull valves for operation;	☐	☐	
b.	Start and stop the Sea Water pump remotely from the MCT;	☐	☐	
c.	Operate the fresh water pumps in remote, explain purpose of standby setting;	☐	☐	
5.	**Operate the Chilled Water Cooling System**			
a.	Properly line up the chilled water system for operation;	☐	☐	
b.	Start and stop the chillers;	☐	☐	
c.	Operate the manual dump valve to achieve proper load on the chiller;	☐	☐	

Item	EC 001.02, 003.01 – 003.03 AND 004.01 Conduct Engineering Rounds	SAT	UNSAT	Remarks
6. Operate the Ventilation System				
a.	Drain down and open up the induction system;	☐	☐	
b.	Drain down and open up the induction system;	☐	☐	
c.	Operate the ship's ventilation in local and remote.	☐	☐	
Score: / The trainee must achieve 6/9 SATs to pass EC 003.01				
7. Operate the Main and External Hydraulic Systems				
a.	Raise Comms mast from all locations;	☐	☐	
b.	State where to collapse main and external hydraulic systems and plants;	☐	☐	
c.	Operate main and external plants from all locations	☐	☐	
8. Operate Steering and Hydroplanes Hydraulics				
a.	Operate of Steering and Hydroplanes hydraulics in all modes;	☐	☐	
b.	Operate an RSU in hand, observing all precautions;	☐	☐	
c.	Operate the rudder using Local Rate, observing all precautions;	☐	☐	
d.	Collapse Steering and Hydroplanes hydraulics.	☐	☐	
Score: / The trainee must achieve 5/7 SATs to pass EC 003.02				
9. Operate the HP Air compressors				
a.	Operate the GIVs from all positions;	☐	☐	
b.	Conduct rounds of all Air Cross-connections;	☐	☐	
c.	Isolate the after planes air emergency bottle;	☐	☐	
10. Operate Vents and Blows				
a.	Operate Main and Emergency Blows from all locations;	☐	☐	
b.	Operate main vents from all locations;	☐	☐	
11. Operate AV&B				
a.	Identify AV&B Reducers, and state the pressure.	☐	☐	

Item	EC 001.02, 003.01 – 003.03 AND 004.01 Conduct Engineering Rounds	SAT	UNSAT	Remarks
b.	Put a blow on starboard after trim tank.	☐	☐	

Score: /

The trainee must achieve 5/7 SATs to pass EC 003.03

12.	**Operate Grade A, Grade B, and Flood alarms**			
a.	Carry out function test of Grade A alarms;	☐	☐	
b.	Carry out function test of Grade B alarms;	☐	☐	
c.	Carry out function test of Flood alarms.	☐	☐	
13.	**Operate the FDS**			
a.	Inhibit a sensor using the FDS panel;	☐	☐	
b.	Acknowledge alarms in all modes;	☐	☐	
c.	Reset alarmed sensor.	☐	☐	
14.	**Operate the Surveillance**			
a.	Locate all major components within surveillance;	☐	☐	
b.	Build an Operator Assist page;	☐	☐	
c.	Inhibit/enable channels.	☐	☐	

Score: /

The trainee must achieve 6/9 SATs to pass EC 004.01.

Positive Feedback	Improvement Feedback

EC 001.02, 003.01 – 003.03, and 004.01– Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 001.02, 003.01 – 003.03, and 004.01with the candidate.	**Candidate** I have been debriefed on the result of EC 001.02, 003.01 – 003.03, and 004.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 001 MAINTAIN SUBMARINE READINESS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine

Arrangements and Procedures

2. PC 001 will be a practical assessment based on content from EO's 001.01 – 001.02.

3. To pass this PC, the trainee must complete all Requirements as laid out in Annex E – OJPR.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

5. PC 001 will be a practical assessment based on content from EO's 001.01 – 001.02.

6. During this practical assessment, you will be required complete all requirements listed in Annex E – OJPR.

7. References will be provided.

8. Do you have any questions?

PART III – PERFORMANCE CHECK NOTES

PO 002 Operate Propulsion and Generation Systems

EO 002.01 Operate the Diesel Generator

1. **Performance**: Operate the Diesel Generator

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Static diesel trainer;
      (4) MCT
      (5) Lesson plan; and
      (6) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) SCT;
      (3) MCT;
      (4) Victoria diesel trainer.

3. **Standard**: The trainee will operate the diesel generator IAW refs, to include:

   a. Preparing the diesel-electric systems for starting/stopping, IAW refs A5 and B7;

   b. Starting/Stopping and monitoring correct local and remote operation of diesel-electric systems, IAW refs A5 and B7; and

   c. Conduct Emergency Operating Procedures (EOP's), IAW ref A1.

4. **Time Required**:

   a. EO – 66 x 50 min periods; and

   b. EC – 6 x 50 min periods (# of students x50 min).

5. **Instructional Methods**:

   a. Interactive lecture; and

   b. EDI.

6. **Training Aids**:

   a. Ngrain videos;

   b. Statistic Diesel Trainers; and

c. Engine educational video.

7. **Test Details**: This EO will be tested as a practical assessment. Trainee must achieve a minimum mark of 23/32 SATs on non-critical factors **AND** all critical on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: The students will be divided into 2 groups for the MCT/SCT phase. Half the students going to the MCT and half going to the SCT. There will be concurrent runs taking place in both trainers.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Prepare the diesel-electric systems for starting/stopping						
1) Preparing diesel systems, to include: a) Ship's fuel system, b) Engine fuel system, c) Air inlet and engine exhaust, d) Diesel SW cooling, e) Diesel FW cooling, f) Engine lube oil, g) Engine air system.	**650**			A5: SOP 101-105, 109, 111; B7	Interactive Lecture Training Aids: - ngrain Videos; and - Statistic Diesel Trainer	Communication between stations is essential to all parts of operation. (SCOOW, SCC1, MCC2)
Start/Stop and monitor correct local and remote operation of diesel-electric systems						
2) Starting, stopping and monitoring the operation of diesel engine, to include: a) Engine logic, b) Procedure, and c) Faults.	**50**	**100**	**1500**	A5: SOP 101, 106 and 107;	EDI Training Aids: Educational video: starting and stopping the engine for snort generating to include remote and local start (process and team work)	The imitation phase require 2 instructors. The group will be split in two syndicates, training time is indicated IAW. Demo is in the MCT

Conduct Emergency Operating Procedures (EOP's)						
3) Conduct Victoria class EOP's, to include: a) Fire based EOP's (not including hydraulics); b) Flood based EOP's.	**50**	**50**	**900**	A1	EDI	Demo and Imitation to be completed in SCT.

Enabling Check – EC 002.01 Operate Diesel Generator

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Machinery Control Trainer/Ship Control Trainer.

Arrangements and Procedures

2. EC 002.01 will be a practical assessment based on content from EO 002.01

3. To pass this EC, the trainee must achieve 23/32 SATs on non-critical factors **AND** all critical factors to pass this EC.

4. Time required: 2 x 50 min period(s) per trainee (# total (time per trainee x number of trainees)), this will be divided up between 1 period in the MCT, and 1 period in the SCT.

5. References will/will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 002.01 will be a practical assessment based on content from EO 002.01.

7. During this practical assessment, you will be required to Operate Diesel Generator. You must pass this EC to pass EO 002.01.

8. References will/will not be provided.

9. Trainee must achieve 23/32 SATs on non-critical factors **AND** all critical factors to pass this EC IAW ECL 002.01.

10. This assessment will be divided up between 2 periods, 1 in the MCT and 1 in the SCT.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

This assessment will be accomplished in two parts; one part in the MCT, and one part in the SCT. Both parts will be combined to show trainees assessment.

PART IV – ENABLING CHECK LIST

Item	EC 002.01 Operate Diesel Generator	SAT	UNSAT	Remarks
1.	**Preparing the Engine to Start (MCT)**			
a.	**Communication (SCOOW, SCC1, MCC2)**	☐	☐	**Critical**
b.	Line up fuel system;	☐	☐	
c.	Line up diesel sea water system;	☐	☐	
d.	Line up diesel exhaust system;	☐	☐	

Item	EC 002.01 Operate Diesel Generator	SAT	UNSAT	Remarks
e.	Line up diesel lube oil system;	☐	☐	
f.	Line up air start system;	☐	☐	
g.	Compare with SCC:	☐	☐	
	i. Depth gauges ii. Battery ventilation flow meters iii. Stop snorting alarm			
h.	Set engine to standby;	☐	☐	
2.	**Preparing the Engine to Start (SCT)**			
a.	**Communication (SCOOW, SCC1, MCC2)**	☐	☐	**Critical**
b.	With SCC2, raise masts and drain down the induction system;	☐	☐	
c.	Compare with MCC:	☐	☐	
	(1) Depth gauges (2) Battery ventilation flow meters (3) Stop snorting alarm			
d.	Upon start of the engine, compensate for induction and exhaust system;	☐	☐	
3.	**Start the Diesel Engine (MCT)**			
a.	**Communication (SCOOW, SCC1, MCC2)**	☐	☐	**Critical**
b.	Sprag the engine	☐	☐	
c.	Start the engine;	☐	☐	
d.	**Ensure correct operation of all systems;**	☐	☐	**Critical**
e.	Reset lube oil, air start, and alarms;	☐	☐	
f.	Log engine start.	☐	☐	
4.	**Stop the Diesel Engine (MCT)**			
a.	Communication (SCOOW, SCC1, MCC2)	☐	☐	
b.	**Allow exhaust temps to settle on one engine;**	☐	☐	**Critical**
c.	Stop engine;	☐	☐	
d.	**Ensure engine stops correctly; engine logic functions properly;**	☐	☐	**Critical**
e.	Revert engine and engine systems to standby position;	☐	☐	
f.	Log engine stop.	☐	☐	
5.	**Stop the Diesel Engine (SCC)**			
a.	Push the stop snorting alarm and switch telegraphs to stop;	☐	☐	

Item	EC 002.01 Operate Diesel Generator	SAT	UNSAT	Remarks
	b. Shut induction hull valve and snort drain 1 valve;	☐	☐	**Critical**
	c. Lower masts;	☐	☐	
	d. Compensate for water in Induction and Exhaust masts,	☐	☐	
6.	**Conduct EOP Actions (MCT)**			
	a. Carry out immediate actions IAW VCEOP's	☐	☐	**Critical**
	b. Carry out follow up actions as directed to return affected systems to operation.	☐	☐	
7.	**Conduct EOP Actions (SCT)**			
	a. Carry out immediate actions IAW VCEOP's	☐	☐	**Critical**
	b. Carry out follow up actions as directed to return affected systems to operation.	☐	☐	

Score: /

Trainee must achieve 23/32 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 002.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002.01 with the candidate.		**Candidate** I have been debriefed on the result of EC 002.01	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 Operate Propulsion and Generation Systems

EO 002.02 Operate the Main Battery

1. **Performance**: Operate the main battery

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Demo battery cell;
      (4) Lesson Plan; and
      (5) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) MCT; and
      (3) Electrical trainer cell.

3. **Standard**: The trainee will operate the main battery IAW refs A5, B4, and B25, by:

   a. Maintaining state boards, IAW refs; and

   b. Maintaining the battery log, IAW refs.

4. **Time Required**:

   a. EO – 29 x 50 min periods; and

   b. EC – 3 x 50 min periods.

5. **Instructional Methods**:

   a. Interactive lecture; and

   b. EDI.

6. **Training Aids**:

   a. Trainers; and

   b. Student log booklet.

7. **Test Details:** This EO will be tested as a practical assessment. Trainee must achieve 7/9 SATs on the practical EC **AND** all critical on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: NA

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operating the main battery						
1) Operating the main battery to include: a) Battery safety, b) Battery construction, c) Battery cell construction, d) Battery charging procedure, i) Types of charges, ii) Charging rates; and iii) Charging modes. 2) Operating battery associated equipment, to include a) Battery ventilation, b) Battery agitation, c) Battery cooling, d) Battery top-up, e) Battery sumps,	**100**	**50**	**1250**	A5, B4, B25	Interactive Lecture Demo: in trainer (battery)	Teaching Points 1 to 3 are part of the same lesson
Maintaining battery log to include maintain state boards						
3) Maintaining battery log, to include: a) Battery readings b) Properly filling out log, 4) Maintaining battery stateboard.	**10**	**15**	**25**	B25	EDI Training Aids: Student log booklet	Teaching Points 1 to 3 are part of the same lesson Students need to log battery routine for the complete length of this lesson. They should continue practicing this for the remaining of the course.

Enabling Check – EC 002.02 Operate the Main Battery

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Machinery Control Trainer.

Arrangements and Procedures

2. EC 002.02 will be a practical assessment based on content from EO 002.02

3. To pass this EC, the trainee must achieve 7/9 SATs.

4. Time required: 1/2 x 50 min period(s) per trainee (# total (time per trainee x number of trainees)).

5. References will/will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 002.02 will be a practical assessment based on content from EO 002.02

7. During this practical assessment, you will be required to Operate the Main Battery. You must pass this EC to pass EO 002.02.

8. References will/will not be provided.

9. Pass mark 7/9 SATs including all Critical IAW ECL 002.02.

10. You will have 1/2 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

Item	EC 002.02 Operate the Main Battery	SAT	UNSAT	Remarks
1.	**Charge the Main Battery**			
a.	Communication (SCOOW, SCC1, MCC2);	☐	☐	
	i. Ensure specific gravity readings are taken prior to charge.			**Critical**
b.	Close the DG power breakers;	☐	☐	
c.	Excite the power breakers to start power flow to the battery;	☐	☐	
d.	Monitor the voltage/current and follow proper charging procedures;	☐	☐	

Item	EC 002.02 Operate the Main Battery	SAT	UNSAT	Remarks
e.	Ensure proper battery ventilation;	☐	☐	
f.	Ensure the battery maintains the proper temperature using cooling when required;	☐	☐	
g.	Upon reaching initial, inform the SCOOW;	☐	☐	
h.	Upon completion of charge, take the battery off load	☐	☐	
i.	Properly fill out the battery log.	☐	☐	

Score: /9

The trainee must achieve 7/9 SATs including all Criticals to pass this EC.

Positive Feedback	Improvement Feedback

EC 002.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 002.02.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 Operate Propulsion and Generation Systems

EO 002.03 Operate the Main Motor

1. **Performance**: Operate the Main Motor

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Propulsion switchboard trainer;
      (4) Lesson Plan; and
      (5) References.

   b. Denied: nil

   c. Environment:

      (1) Classroom;
      (2) MCT.

3. **Standard**: The trainee will operate the main motor, IAW refs A5 and B8, to include operating the shaftline IAW refs A5, B9, B13 and C1.

4. **Time Required**:

   a. EO – 42 x 50 min periods;

   b. EC – 3 x 50 min periods (# of students x50 min); and

   c. PC – 7 x 50 min periods (# of students x50 min).

5. **Instructional Methods**:

   a. EDI; and

   b. Interactive lecture.

6. **Training Aids**:

   a. Shaftline slide-show / educational video; and

   b. Trainers.

7. **Test Details:** This EO will be tested as a practical assessment. Trainee must attain a minimum mark Trainee must achieve 11/15 SATs **AND** all critical on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: There will be concurrent runs being taking place in both trainers.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate main motor						
1) Operating the main motor, to include: a) Function, b) Construction, to include: i) Propulsion switchboard, ii) Automatic Control Unit (ACU), iii) Excitation Failure Relay Unit (EFRU), iv) Slow speed Motor Generator, c) Operation, i) Modes of operation, ii) Parameters, iii) Speed, iv) Power Ratings,	**500**	**150**	**1300**	A5: SOP 120, B8	Interactive Lecture Training aids: Slide show (pictures), Propulsion Switchboard Trainer.	The imitation phase require 2 instructors. The group will be split in two syndicates, training time is indicated IAW.
Operate the Shaftline						
2) Operating the shaftline, to include: a) Main motor bearings, i) Construction, ii) Operation, b) Main shaft, to include: i) Construction, ii) Operation, c) Thrust block, to include: i) Construction, ii) Operation, d) Shaft seal, to include: i) Construction, ii) Operation,	**150**			A5, B9, B13 and C1	Interactive Lecture Training aids: Slide-show (pictures) S1: Animated video (how the thrust block, turning gear, shaft seal and pilgrim nut work).	

e) Propeller, to include: i) Construction, ii) Operation, iii) Pilgrim nut. f) Turning gear and shaft lock, to include: i) Operation.						

Enabling Check – EO 002.03 Operate the Main Motor

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Machinery Control Trainer.

Arrangements and Procedures

2. EC 002.03 will be a practical assessment based on content from EO 002.03.

3. To pass this EC, the trainee must achieve 11/15 SATs.

4. Time required: 1/2 x 50 min period(s) per trainee (3 total (time per trainee x number of trainees)).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 002.03 will be a practical assessment based on content from EO 002.03

7. During this practical assessment, you will be required to Operate the Main Motor. You must pass this EC to pass EO 002.03.

8. References will not be provided.

9. Pass mark is 11/15 SATs IAW ECL 002.03.

10. You will have 1/2 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

Item	EC 002.03 Operate the Main Motor	SAT	UNSAT	Remarks
1.	**Operate the Main Motor**			
	a. Switch mode of operation to Manual Mode 1;	☐	☐	
	b. Respond to changes in group, by:			
	i. Increasing to full field;	☐	☐	
	ii. Switch starting cam to off;	☐	☐	
	iii. Witch grouping cam to desired group;	☐	☐	
	iv. Switch starting cam to correct direction;	☐	☐	
	v. Adjust field current to required field amperage.	☐	☐	
2.	**Respond to Propulsion Failure – Loss of Field Converter**			
	a. Switch mode of operation to Manual Mode 1;	☐	☐	
	c. Switch field current to full field;	☐	☐	
	d. Wind starting and grouping camshafts to stop and off;	☐	☐	
	e. Open both main motor breakers;	☐	☐	**Critical**
	f. Switch the failed field converter to off;	☐	☐	
	g. Set the affected field selector switch to standby;	☐	☐	
	h. Close both main motor breakers;	☐	☐	
	i. Ensure proper operation of the field current and cam shaft drive motors;	☐	☐	
	j. Restore propulsion to Primary Control.	☐	☐	

Score: /15

The trainee must achieve 11/15 SATs, including all Critical Factors, to pass this EC.

Positive Feedback	Improvement Feedback

Item	EC 002.03 Operate the Main Motor	SAT	UNSAT	Remarks

EC 002.03 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002.03 with the candidate.		**Candidate** I have been debriefed on the result of EC 002.03.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 002 OPERATE PROPULSION AND GENERATOR SYSTEMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. MCT

Arrangements and Procedures

2. PC 002 will be a practical assessment based on content from EO's 002.01 – 002.03.

3. To pass this PC, the trainee must complete 21/29 SATs including all criticals.

4. Time required: 1 x 50 min period per trainee (6 period's total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. PC 002 will be a practical assessment based on content from EOs 002.01 – 002.03.

7. During this practical assessment, you will be required to start the engines, put them on load and respond to a Propulsion Failure. You must pass this PC to pass EOs 002.01 – 002.03.

8. References will not be provided.

9. Pass mark is 21/29 SATs, including all Criticals, IAW PCL 001.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

Item	PC 002 Performance Statement	SAT	UNSAT	Remarks
1.	**Prepare the diesel generator to start.**			
a.	Communication (SCOOW, SCC1, MCC2)	☐	☐	**Critical**
b.	Line up fuel system;	☐	☐	
c.	Line up diesel sea water system;	☐	☐	
d.	Line up diesel exhaust system;	☐	☐	
e.	Line up diesel lube oil system;	☐	☐	
f.	Line up air start system;	☐	☐	
g.	Compare with SCC:	☐	☐	
	i.Depth gauges ii.Battery ventilation flow meters iii.Stop snorting alarm			
h.	Set engine to standby;	☐	☐	
2.	**Start the Diesel Engine (MCT)**			
a.	**Communication (SCOOW, SCC1, MCC2)**	☐	☐	
b.	Sprag the Engine	☐	☐	
c.	Start the engine;	☐	☐	
d.	Ensure correct operation of all systems;	☐	☐	
e.	Reset lube oil, air start and alarms;	☐	☐	
f.	Log engine start.	☐	☐	
3.	**Charge the Main Battery**			
a.	Communication (SCOOW, SCC1, MCC2);	☐	☐	
	i. Ensure specific gravity readings are taken prior to charge.			**Critical**
b.	Close the DG power breakers;	☐	☐	
c.	Excite the power breakers to start power flow to the battery;	☐	☐	
d.	Monitor the voltage/current and follow proper charging procedures;	☐	☐	
e.	Ensure proper battery ventilation;	☐	☐	

Item	PC 002 Performance Statement	SAT	UNSAT	Remarks
	f. Ensure the battery maintains the proper temperature using cooling when required;	☐	☐	
4.	**Respond to Propulsion Failure – Loss of ACU**			
	a. Stop engines;	☐	☐	
	b. Open DG breakers;	☐	☐	
	c. Secure the air charge;	☐	☐	
	d. Switch mode of operation to Manual Mode 1;	☐	☐	
	e. Switch field current to full field;	☐	☐	
	f. Match Telegraphs;	☐	☐	
	g. Switch the ACU to off;	☐	☐	
	h. Ensure proper operation of the field current and cam shaft drive motors;	☐	☐	
	i. Respond to telegraphed propulsion orders.	☐	☐	

Score: /29

The trainee must achieve 21/29 SATs, including all Critical Factors, to pass this PC.

Positive Feedback	Improvement Feedback

PC 002 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of PC 002 with the candidate.	**Candidate** I have been debriefed on the result of PC 002.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 Operate Main Mechanical Systems

EO 003.01 Operate the Cooling Water Systems

1. **Performance**: Operate the cooling water systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: nil

   c. Environment:

      (1) Classroom;
      (2) SCT; and
      (3) MCT;

3. **Standard**: The trainee will operate the cooling water systems to include:

   a. Operating the sea water cooling system, IAW refs A5 and B15;
   b. Operating the fresh water cooling system, IAW refs A5 and B15;
   c. Operating the chilled water cooling system, IAW refs A5 and B12, to include operating ventilation systems, IAW A5 and B11.

4. **Time Required**:

   a. EO – 5 x 50 min periods; and

5. **Instructional Methods**:

   a. Field Trip;
   b. Interactive lecture; and
   c. EDI.

6. **Training Aids**: Nil.

7. **Test Details:** This EO will be tested as part of EC 001.02 as a practical test, this will concurrently test EO's 001.02, 003.01 – 003.03, and 004.01 and will occur during a tour of an available submarine. In the event that a submarine is unavailable, the assessment will be written. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**:

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the sea water and fresh water cooling system						
1) Operating the sea water cooling, to include: a) Function, b) Major components, to include: i) SW Pumps, ii) Heat Exchangers, iii) Hull Valves, iv) Equipment Cooled, c) Operation, to include: i) System parameters. 2) Operating the fresh water cooling, to include: a) Function, b) Major components, to include: i) FW Pumps, ii) Heat Exchangers, iii) Hull Valves, iv) Equipment Cooled c) Operation, to include: i) System parameters	**50**	**10**	**15**	A5: SOP 230 and 233	EDI (incl. Field Trip)	
Operate the chilled water cooling system						
3) Operating the AC chiller, to include: a) Function of plant, b) Major components, c) "Fridge Cycle" d) System operation 4) Operating the chilled water system, to include: a) Function, b) Services fed, c) Alternate operation, to include: i) Ship supplied, ii) Shore Supplied.	**50**	**10**	**10**	A5: SOP 231 and 232	EDI (incl. Field Trip)	
Operate ventilation systems						
5) Operating the ventilation systems, to include: a) Induction system i) Function, ii) Components,	**50**	**10**		A5: SOP 223, 226 and 227.	EDI (incl. Field Trip)	

iii) Operation b) VEX system i) Function, ii) Components, iii) Operation c) Ship's ventilation system i) Function, ii) Components, iii) Operation						

Enabling Check – EO 003.01 Conduct Engineering Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine, field trip.

Arrangements and Procedures

2. EC 003.01 will be a practical assessment based on content from EO 003.01. It will be conducted during a field trip to the submarine concurrently with ECs 001.02, 003.01 – 003.03, and 004.01. The student will be required to demonstrate their abilities to conduct pre-watch and on watch rounds to the instructor.

3. To pass this EC, the trainee must achieve the following:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

4. Time required: 1 x 50 min period(s) per trainee (6 total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. This walkthrough will be a practical assessment based on content from EO 001.02, 003.01 – 003.03, and 004.01.

7. During this practical assessment, you will be required to conduct Engineering Rounds. You must pass this EC to pass EO 001.02, 003.01 – 003.03, and 004.01.

8. References will not be provided.

9. To pass this EC, the trainee must achieve the following, IAW ECL 001.02:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PO 003 Operate Main Mechanical Systems

EO 003.02 Operate the Hydraulic Systems

1. **Performance**: Operate the hydraulic systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: nil

   c. Environment:

      (1) Classroom; and
      (2) SCT.

3. **Standard**: The trainee will operate the hydraulic systems to include:

   a. Operating the main, external, and steering and hydroplanes plants IAW refs A5 and B13;

   b. Operating the main and external hydraulic systems IAW refs A5 and B13;

   c. Operating steering and hydroplanes IAW refs A5 and B13; and

   d. Performing a Bathythermograph, IAW refs A10.

4. **Time Required**:

   a. EO – 5 x 50 min periods;

5. **Instructional Methods**:

   a. EDI; and

   b. Interactive lecture.

6. **Training Aids**: nil.

7. **Test Details**: This EO will be tested as part of EC 001.02 as a practical test, this will concurrently test EO's 001.02, 003.01 – 003.03, and 004.01 and will occur during a tour of an available submarine. In the event that a submarine is unavailable, the assessment will be written. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the main, external, and steering and hydroplanes plants						
1) Operating the plants to include: a) Function and Operation b) Raft Mounted Equipment c) Rep Tank Function/Capacity d) Modes of Operation, to include: i) PD Line-up ii) Deep Line-up e) Alarms and Protection, to include i) Fire Isolation Valves (FIV) ii) Grade A&B Alarms iii) Reliefs f) Plant Accumulators, to include: i) Purpose, ii) Burst Bag routine. g) Hydraulic burst EOP;	**25**			A5: SOP 201, 202 and 203	EDI (React with EOP)	
Operate the main and external hydraulic systems						
2) Operating the main and external hydraulic components, to include: a) Hull valves and system accumulators, b) Mast operation, c) Collapsing system hydraulics,	**25**	**25**		A5: SOP 201 and 202	Interactive Lecture	Taught with TP 1
Operate steering and hydroplanes						
3) Operating the steering and hydroplanes system, to include: a) Rudder, to include: i) System operation, ii) Modes of operation, b) Afterplanes, i) System operation,	**50**	**15**	**35**	A5: SOP 203	EDI (React with EOP) Field Trip	Demo and Prac. Local Operation of Steer and Hydroplanes on Submarine

ii) Modes of operation c) Foreplanes, i) System operation, ii) Modes of operation d) Loss of control surface EOP.						
Perform a Bathythermograph						
4) Operating the After SSE, to include: a) Loading procedure, b) Firing procedure.	**50**	**25**		A10	EDI	

Enabling Check – EO 003.02 Conduct Engineering Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine, field trip.

Arrangements and Procedures

2. EC 003.02 will be a practical assessment based on content from EO 003.02. It will be conducted during a field trip to the submarine concurrently with ECs 001.02, 003.01 – 003.03, and 004.01. The student will be required to demonstrate their abilities to conduct pre-watch and on watch rounds to the instructor.

3. To pass this EC, the trainee must achieve the following:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

4. Time required: 1 x 50 min period(s) per trainee (6 max periods per student (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. This walkthrough will be a practical assessment based on content from EO 001.02, 003.01 – 003.03, and 004.01.

7. During this practical assessment, you will be required to conduct Engineering Rounds. You must pass this EC to pass EO 001.02, 003.01 – 003.03, and 004.01.

8. References will not be provided.

9. To pass this EC, the trainee must achieve the following, IAW ECL 001.02:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PO 003 Operate Main Mechanical Systems

EO 003.03 Operate the Air Systems

1. **Performance**: Operate the air systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) SCT; and
      (3) MCT.

3. **Standard**: The trainee will operate air systems, IAW refs A5 and B10, to include carrying out vacuum test, IAW refs A5.

4. **Time Required**:

   a. EO – 11 x 50 min periods; and

5. **Instructional Methods**:

   a. EDI; and

   b. Interactive lecture.

6. **Training Aids**: Nil.

7. **Test Details**: This EO will be tested as part of EC 001.02 as a practical test, this will concurrently test EO's 001.02, 003.01 – 003.03, and 004.01 and will occur during a tour of an available submarine. In the event that a submarine is unavailable, the assessment will be written. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**:

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the HP air system						
1) Operating the HP air compressors to, to include: a) Construction; b) Pre-start Checks; c) Start-up/Shut-down Procedures; 2) Operating the HP air system parameters, to include: a) Bottle Groups; b) Ring Mains; c) Cross Connections; d) System Isolation/Integrity; e) HP air burst EOP.	**25**	**25**	**25**	A5: SOP 219 – 221; B10	Interactive Lecture/Field Trip Training Aids: Slide show presentation	Demo/Practical will Take Place during field trip to Submarine. Operation to be demonstrated by SME during field trip.
3) Operating vents and blows a) Main Vents, to include: i) Function; ii) Operation. b) Main Blows: i) Function; ii) Operation. c) Emergency Blows i) Function ii) Operation	**50**	**50**	**150**	A5: SOP 219 – 221; B10	EDI	Demo/Practical will take place in the SCT
Carrying out vacuum test						
4) Carrying out a vacuum test, to include: a) Purpose, b) Line up, c) Procedure, d) Breaking the vacuum.	**50**	**25**	**125**	A5: SOP 228; B10	EDI (Interactive Demo)	Demo/Practical will take place in the SCT
Operate the Auxiliary Vent and Blow System						
5) Operating the AV&B system to include:	**25**			A5: SOP 222;	Interactive Lecture	

a) System function and construction, to include i) Main Reducers, ii) Secondary Reducers, iii) Other Services. b) Reducer Operation and Setup.				B10		

Enabling Check – EO 003.03 Conduct Engineering Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine, field trip.

Arrangements and Procedures

2. EC 003.03 will be a practical assessment based on content from EO 003.03. It will be conducted during a field trip to the submarine concurrently with ECs 001.02, 003.01 – 003.03, and 004.01. The student will be required to demonstrate their abilities to conduct pre-watch and on watch rounds to the instructor.

3. To pass this EC, the trainee must achieve the following:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

4. Time required: 1 x 50 min period(s) per trainee (6 total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. This walkthrough will be a practical assessment based on content from EO 001.02, 003.01 – 003.03, and 004.01.

7. During this practical assessment, you will be required to conduct Engineering Rounds. You must pass this EC to pass EO 001.02, 003.01 – 003.03, and 004.01.

8. References will not be provided.

9. To pass this EC, the trainee must achieve the following, IAW ECL 001.02:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PO 003 Operate Main Mechanical Systems

EO 003.04 Operate Trim, Bilge and Ballast Systems

1. **Performance**: Operate Trim, Bilge and Ballast Systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: nil

   c. Environment:

      (1) Classroom; and
      (2) SCT.

3. **Standard**: The trainee will operate trim, bilge and ballast systems, IAW refs A5 and B14, to include:

   a. Operating the HP bilge and ballast systems IAW refs;

   b. Operating the trim system IAW refs; and

   c. Operating the LP bilge system IAW refs.

4. **Time Required**:

   a. EO – 38 x 50 min periods;

   b. EC – 6 x 50 min periods; and

   c. PC – OJPR

5. **Instructional Methods**: EDI.

6. **Training Aids**: Equipment (HP bilge, ballast and trim).

7. **Test Details**: This EO will be tested as a practical assessment. Trainee must attain a minimum mark Trainee must achieve 6/8 SATs on non-critical factors **AND** all critical on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the HP bilge and ballast systems						
1) Operating the HP bilge and ballast system, to include: a) Function, b) Design, to include: i) Available tanks, ii) Positive suction requirement, iii) Water Sensitive Probe c) Operation, to include: i) Speed Limitations, ii) Depth limitation, iii) Flow Rate Limitation. 2) Operating the priming System, to include: a) Function, System Operation	**100**	**100**	**1000**	A5: SOP 209, 211 and 212, B14	EDI Demo 1: HP bilge and ballast Demo 2: trim Demo 3: combine performance	The imitation phase require 2 instructors. The group will be split in two syndicates, training time is indicated IAW. Stress the importance of communication as it is evaluated during the EC.
Operate the trim system						
3) Operating the trim system to include: a) Pumps, b) Tanks, c) Operation of the system i) Normal, ii) Alternate d) Restrictions,	**25**	**25**	**500**	A5: SOP 209, 210, B14	TP 1 to 3 taught together	The imitation phase require 2 instructors. The group will be split in two syndicates, training time is indicated IAW. TP 1 to 3 taught together
Operate the LP bilge system						

4) Operating the LP bilge system, to include: a) Pumps, b) Tanks, c) Procedures, d) Restrictions	**25**	**25**	**100**	A5: SOP 209, 212, B14	EDI	The imitation phase require 2 instructors. The group will be split in two syndicates, training time is indicated IAW.

Enabling Check – EC 003.04 Operate Trim, Bilge and Ballast Systems

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT.

Arrangements and Procedures

2. EC 003.04 will be a practical assessment based on content from EO's 003.04.

3. To pass this EC, the trainee must achieve 6/8.

4. Time required: 1 x 50 min period(s) per trainee (6 total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.04 will be a practical assessment based on content from EO 003.04.

6. References will not be provided.

7. Pass mark is 6/8

8. You will have 1 x 50 min period(s) to complete this assessment.

9. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The operator of the SCT will change the state of the trim tanks to affect the boats buoyancy in a positive or negative manner and change the trim forward or aft. The trainee will then have 1 hour to compensate and bring the submarine back into a neutral state.

PART IV – ENABLING CHECK LIST

Item	EC 003.04 Operate the Trim, Bilge and Ballast Systems	SAT	UNSAT	Remarks
1. **Operate Trim, Bilge and Ballast**				
	a. Correct ballast and trim of submarine to achieve a neutral buoyancy;	☐	☐	**Critical**
	b. Communicate with helmsman;	☐	☐	

Item	EC 003.04 Operate the Trim, Bilge and Ballast Systems	SAT	UNSAT	Remarks
c.	Observes limitations and precautions, to include:	☐	☐	
i.	Depth;	☐	☐	
ii.	Speed;	☐	☐	
iii.	Pump parameters;	☐	☐	
iv.	Tank levels;	☐	☐	
v.	Adverse conditions.	☐	☐	

Score: /8

Trainee must achieve 6/8 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 003.04 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 003.04 with the candidate.		**Candidate** I have been debriefed on the result of EC 003.04.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 003 OPERATE MAIN MECHANICAL SYSTEMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine

Arrangements and Procedures

2. PC 003 will be a practical assessment based on content from EO's 003.01 – 003.04.

3. To pass this PC, the trainee must complete all associated Requirements as laid out in Annex E – OJPR.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

5. PC 003 will be a practical assessment based on content from EO's 003.01 – 003.04.

6. During this practical assessment, you will be required complete all associated requirements listed in Annex E – OJPR.

7. References will be provided.

8. Do you have any questions?

PART III – PERFORMANCE CHECK NOTES

PO 004 Operate Electrical Systems

EO 004.01 Operate Warning Systems

1. **Performance**: Operate warning systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Central Surveillance System Trainer;
      (4) Lesson Plan; and
      (5) References.

   b. Denied: nil

   c. Environment:

      (1) Classroom;
      (2) Electrical Training Cell;
      (3) SCT; and
      (4) MCT.

3. **Standard**: The trainee will operate the warning systems to include:

   a. 24V Grade A and B, and Flood alarms, IAW refs A5 and B5;

   b. Fire Detection System, IAW refs C6; and

   c. Central Surveillance System, IAW C4.

4. **Time Required**:

   a. EO – 6 x 50 min periods; and

5. **Instructional Methods**: EDI.

6. **Training Aids**: Submarine.

7. **Test Details**: This EO will be tested as part of EC 001.02 as a practical test, this will concurrently test EO's 001.02, 003.01 – 003.03, and 004.01 and will occur during a tour of an available submarine. In the event that a submarine is unavailable, the assessment will be written. Practical application of this EO will be tested as part of the OJPR. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the warning systems						
1) Operating the Grade A, Grade B and Flood Alarms, to include: a) Purpose; and b) Operation.	**25**			A5 and B5	Interactive Computer-based module	
2) Operating the Fire Detection System, to include: a) Function; b) Location; and c) Operation.	**50**	**25**	**25**	C6	EDI	
3) Operating the CSS, to include: a) System Functions; b) Location of major items, to include: i) VDU's; ii) LSU's; iii) CPU; and iv) POU. c) Modes of Operation; and d) Capabilities.	**75**	**15**	**85**	C4	EDI	

Enabling Check – EO 004.01 Conduct Engineering Rounds

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Submarine, field trip.

Arrangements and Procedures

2. EC 001.02 will be a practical assessment based on content from EO 003.0. It will be conducted during a field trip to the submarine concurrently with ECs 001.02, 003.01 – 003.03, and 004.01. The student will be required to demonstrate their abilities to conduct pre-watch and on watch rounds to the instructor.

3. To pass this EC, the trainee must achieve the following:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

4. Time required: 1 x 50 min period(s) per trainee (6 total (time per trainee x number of trainees)).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. This walkthrough will be a practical assessment based on content from EO 001.02, 003.01 – 003.03, and 004.01.

7. During this practical assessment, you will be required to conduct Engineering Rounds. You must pass this EC to pass EO 001.02, 003.01 – 003.03, and 004.01.

8. References will not be provided.

9. To pass this EC, the trainee must achieve the following, IAW ECL 001.02:
    a. EC 002.02 – 12/17 SATs
    b. EC 003.01 – 6/9 SATs;
    c. EC 003.02 – 5/7 SATs;
    d. EC 003.03 – 5/7 SATs; and
    e. EC 004.01 – 6/9 SATs.

10. You will have 1 x 50 min period(s) to complete this assessment.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PO 004 Operate Electrical Systems

EO 004.02 Operate the Variable Pressure Direct Current (VPDC)

1. **Performance**: Operate the VPDC

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) MCT.

3. **Standard**: The trainee will operate the VPDC, IAW refs A5 and B3, to include:

   a. Power breakers;

   b. Auxiliary DC system; and

   c. Battery switchboard.

4. **Time Required**:

   a. EO – 7 x 50 min periods;

   b. EC – 3 x 50 min periods.

5. **Instructional Methods**:

   a. Field Trip; and

   b. EDI.

6. **Training Aids**: SCT/MCT.

7. **Test Details**: This EO will be tested as a practical assessment. Trainee must attain a minimum mark Trainee must achieve 6/9 SATs on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the VPDC						
1) Operating the battery switchboard, to include: a) Function; b) Power breakers; c) Mid-point services; d) Shore Supply; e) Control fuses; f) Links.	**25**			A5: SOP 125, 126 and 128; B3	Field Trip	On this field trip the objective is to operate the system with SME.
2) Operating the auxiliary DC system, to include: a) Load centres, and b) Services; c) DC failure EOP.	**25**	**50**	**250**	A5: SOP 125; B3	EDI	

Enabling Check – EC 004.02 Operate VPDC

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 004.02 will be a practical assessment based on content from EO 004.02.
3. Trainee must achieve 6/9 SATs to pass this EC.
4. Time required: 1/2 x 50 min period(s) per trainee (3 total (time per trainee x number of trainees))
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. EC 004.02 will be a practical assessment based on content from EO's 004.01 – 004.02.
7. References will not be provided.
8. Trainee must achieve 6/9 SATs to pass this EC.
9. You will have 1/2 x 50 min period(s) to complete this assessment.
10. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

Item	EC 004.02 Operate VPDC System	SAT	UNSAT	Remarks
1.	**(SCT) Respond to VCEOP – Electrical Failure – Loss of DC Load Centre**			
a.	Stop Generating, Ventilating	☐	☐	
b.	Switch Ship`s Vent, Battery Vent, Fuel Firesafe Switched to Emerg Shutdown	☐	☐	
c.	Make appropriate changes to lineup, to include:			
i.	Switch affected pumps to off;	☐	☐	
ii.	Switch Steering and Hydroplane to line;	☐	☐	
iii.	Report to SCOOW machinery losses.	☐	☐	
2.	**(SCT) Respond to VCEOP – Electrical Failure – Loss of 24V DC Grade 2**			
a.	Stop Generating, Ventilating;	☐	☐	
b.	Switch Ship`s Vent, Battery Vent, Fuel Firesafe Switched to Emerg Shutdown;	☐	☐	
c.	Recognize losses;	☐	☐	
d.	Take appropriate actions to restore system functionality.	☐	☐	

Score: /9

Trainee must achieve 6/9 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**

Item	EC 004.02 Operate VPDC System	SAT	UNSAT	Remarks

Assessor I have discussed the results of EC 004.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 004.02	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 004 Operate Electrical Systems

EO 004.03 Operate the Alternating Current (AC) Systems

1. **Performance**: Operate the AC systems

2. **Conditions**:

   a. Given:

      (1) MCAN computer and projector;
      (2) Whiteboard;
      (3) Lesson Plan; and
      (4) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) SCT; and
      (3) MCT.

3. **Standard**: The trainee will operate the AC systems, IAW refs A5 and B6, to include operating the 24 VDC system IAW A5 and B5.

4. **Time Required**:

   a. EO – 19 x 50 min periods;

   b. EC – 3 x 50 min periods; and

   c. PC – 7 x 50 min periods.

5. **Instructional Methods**: EDI.

6. **Training Aids**: MCT/SCT.

7. **Test Details**: This EO will be tested as a practical assessment. Trainee must attain a minimum mark Trainee must achieve 8/12 SATs on non-critical factors **AND** all critical on the practical EC. An onboard assessor will sign off the appropriate block in the OJPR once satisfied with the candidate's performance.

8. **Remarks**: nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Operate the Alternating Current (AC) distribution system						
1) Operating the AC distribution system, to include: a) 140KW MGs b) Shore Supply c) AC Switchboards and load centers; d) Services; e) AC failure EOP.	**100**	**100**	**700**	A5; and B6	EDI	
Operate the 24V DC system						
2) Operating the 24V DC distribution system, to include: a) Transformer Rectifier Units; b) 24V distribution panel; c) LP battery; d) Grade 1 and Grade 2 services; e) Loss of 24V EOP.	**25**	**25**		A5; and B5	EDI	

Enabling Check – EC 004.03 Operate AC Systems

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 004.03 will be a practical assessment based on content from EO 004.03.

3. Trainee must achieve 8/12 SATs on non-critical factors **AND** all critical factors to pass this EC.

   Time required: 1/2 x 50 min period(s) per trainee (3 max periods per student (time per trainee x number of trainees)).

4. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

5. EC 004.03 will be a practical assessment based on content from EO 004.03.

6. References will not be provided.

7. Trainee must achieve 8/12 SATs on non-critical factors **AND** all critical factors to pass this EC.

8. You will have 1 x 50 min period(s) to complete this assessment.

9. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

Item	EC 004.03 OPERATE AC SYSTEMS	SAT	UNSAT	Remarks
1. **Transfer from Shore supply to MG supply**				
a.	Line up MG for correct running;	☐	☐	**Critical**
b.	Make appropriate warning pipes;	☐	☐	**Critical**
c.	Start the MG;	☐	☐	
d.	Check for earths;	☐	☐	**Critical**
e.	Synchronize with shore supply;	☐	☐	

Item	EC 004.03 OPERATE AC SYSTEMS	SAT	UNSAT	Remarks
f.	Connect MG to Bus.	☐	☐	
2.	**Connect second MG to Bus**			
a.	Line up MG for correct running;	☐	☐	**Critical**
b.	Make appropriate warning pipes;	☐	☐	**Critical**
c.	Start the MG;	☐	☐	
d.	Check for earths;	☐	☐	**Critical**
e.	Synchronize with running MG;	☐	☐	
f.	Connect second MG to Bus.	☐	☐	

Score: /12

Trainee must achieve 8/12 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.03 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 004.03 with the candidate.		**Candidate** I have been debriefed on the result of EC 004.03.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 004 OPERATE ELECTRICAL SYSTEMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. MCT

Arrangements and Procedures

2. PC 004 will be a practical assessment based on content from EO's 004.02-004.03.
3. Trainee must achieve 17/21 SATs on non-critical factors **AND** all critical factors to pass this EC
4. Time required: 1 x 50 min period per trainee (6 max periods per student (time per trainee x number of trainees)).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. PC 004 will be a practical assessment based on content from EO's 004.02 – 004.03.
7. During this practical assessment, you will be required to Operate the AC and VPDC systems. You must pass this PC to pass PO 004.
8. References will not be provided.
9. Trainee must achieve 17/21 SATs on non-critical factors **AND** all critical factors to pass this EC IAW PCL 001.
10. You will have 1 x 50 min period(s) to complete this assessment.
11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Give guidance for the development of the assessment.

PART IV – PERFORMANCE CHECK LIST

Item	PC 004 Performance Statement	SAT	UNSAT	Remarks
1.	**React to VCEOP Emergency Stations – Fire in the Main Battery**			
a. b.	Stop Snorting, Generating, Ventilating; Secure the air charge; Man propulsion switchboard in Man Mode 1, if required.	☐	☐	
c.	Make appropriate warning pipes;	☐	☐	
d.	Open affected MG Breaker; Shut down affected MG; switch to alternate supply;	☐	☐	
e.	Open affected DC LC breaker;	☐	☐	
f.	Open appropriate Aux Power Breaker;	☐	☐	
g.	**Open Appropriate Main Power Breaker;**	☐	☐	**Critical**
h.	Isolate PV valves;	☐	☐	
i.	Make appropriate warning pipes;	☐	☐	
j.	Make Alt. DC LC breaker;	☐	☐	
k.	Check if MG is run down;	☐	☐	
l.	Once run down, make appropriate pipes and start MG in alternate supply	☐	☐	
m.	Once MG is checked correct, make appropriate pipes, connect MG to bus;	☐	☐	
n.	Report that power has been restored.	☐	☐	
2.	**Revert power to normal supply**			
a.	**Ensure a specific gravity readings are taken and are within parameters;**	☐	☐	**Critical**
b.	**Make appropriate warning pipes;**	☐	☐	**Critical**
c.	Close Main and Aux power breakers;	☐	☐	
d.	Switch DC LC to normal supply;	☐	☐	
e.	Synchronize MG for single MG running;	☐	☐	

Item	PC 004 Performance Statement	SAT	UNSAT	Remarks
f.	Shut down MG and revert to normal supply; start MG following all precautions;	☐	☐	
g.	Revert AC supplies to SOP line up;	☐	☐	
h.	Report power is in SOP lineup.	☐	☐	

Score: /21

Trainee must achieve 7/21 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

PC 004 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of PC 004 with the candidate.	**Candidate** I have been debriefed on the result of PC 004.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Code	Document Number	Publication
A1		VCEOP's
A2		NAVORD's
A3		NEM
A4		VCSSO's
A5		VCSOP's Vol. 1 &2
A6		Victoria Class SEMS
A7	C-23-VIC-000	Quality Assurance for First Level Systems and Areas
A8	MARPACORD G-108	MARPAC Lockout/Tagout Program
A9	MARLANTORD 66-3	MARLANT Lockout/Tagout Program
A10	C-69-820-000/MB-008	SUBWOPS Vol. 3
B1	BR 1966(2)	Firefighting Systems & Equipments
B2	BR 1966(3)	Escape and Salvage Systems and Equipment
B3	BR 1966(4)	VPDC Systems & Equipment – Index
B4	BR 1966(5)	Main Battery Systems & Equipment – Index
B5	BR 1966(6)	24 VDC Systems & Equipment – Index
B6	BR 1966(7)	AC Systems & Equipment – Index
B7	BR 1966(8)	Diesel Engines, Installation, Systems, & Equipment – Index
B8	BR 1966(9)	Prop: Electrical Systems & Equipment – Index
B9	BR 1966(10)	Prop: Mechanical Systems & Equipment – Index
B10	BR 1966(11)	Compressed Air & Nitrogen Systems & Equipment – Index
B11	BR 1966(12)	Ventilation, Air Conditioning & Snort Induction – Index
B12	BR 1966(13)	Chilled Water Systems & Equipment – Index
B13	BR 1966(14)	Ships Hydraulic Systems & Equipment – Index
B14	BR 1966(15)	Trim, Bilge & Ballast Systems
B15	BR 1966(16)	Ships SW System & Equipment
B16	BR 1966(17)	Ships & Domestic FW System & Equipment
B17	BR 1966(18)	Desalination Plant
B18	BR 1966(20)	Surveillance System and Equipment
B19	BR 1966(21)	Control Consoles – Index
B20	BR 1966(22)	Rudder & Hydroplanes, Control System & Equip – Index
B21	BR 1966(23)	Masts & Associated Equipment – Index
B22	BR 1966(24)	Ship's Stores Refrigeration
B23	BR 1966(25)	Domestic Equipment
B24	BR 1966(30)	Lighting Systems & Equipment – Index
B25	BR 2075(K)	Submarine Main Batteries
B26	BR 3623(01)	TAB's – Ships Systems
B27	BR 3623(02)	TAB's – Propulsion
B28	BR 3623(03)	TAB's – Tactical Weapons Systems
B29	BR 3623(04)	TAB's – Electrical Systems
B30	BR 4500(526)	Static Frequency Changer, MK38SFC
B31	BR 4500(529)	Static Frequency Changer, MK36SFC
B32	BR 4500(701)	Batteries, Primary & Secondary
B33	BR 4569	Pressure & Differential Pressure Transducers & Associated Equipment
B34	BR 6528(301)	Paxman Valenta Diesel Engine 16 RPA 200z
B35	BR 6528(303)	Valenta Engines Dockyard Overhaul/Repair
B36	BR 8223	Breathing Apparatus
C1		Propulsion Power Statement

Code	Document Number	Publication
C2		Composite Drawings
C3		Autopilot Manual
C4		CSS Manual
C5	DRMIS-SIGRD.mil.ca/index.eng.asp	DRMIS Home Page
D1		

A CF or other official Canadian government publication
B Official Allied Force or Allied government publication
C Commercial publication
D Other

Number	Task	T/NT	PO
DT0200	Operate main seawater circulation system remotely	T	003
JT0260	Operate fresh water system locally	T	003
YT0090	Operate chilled water system (subs)	T	003
YT0110	Operate hydraulic Plant remotely (subs)	T	003
PT0310	Operate propulsion switchboards	T	002
AUT1250	Inspect main battery charging system (electric propulsion)	T	004
YT0120	Operate main ballast tank high pressure blow system remotely (Sub)	T	003
YT0140	Operate main ballast tank high pressure blow system locally (Sub)	T	003
YT0180	Operate ship's ventilation systems locally (subs)	T	003
YT0190	Operate ship's ventilation systems remotely (subs)	T	003
YT0200	Operate submarine battery ventilation system locally	T	002
YT0070	Operate ballast system (sub)	T	003
YT0050	Maintain submarine trim	T	003
YT0220	Operate surveillance system remotely (Sub)	T	004
YT0210	Operate surveillance system locally (sub)	T	004
AKT1090	Investigate system operation logbooks	T	001
AUT4090	Update battery log	T	002
ZT0020	Conduct engineering pre-sailing checks	T	001
IT 0470	Manage propulsion plant operation in emergency and action stations	T	002
AFT0400	Report maintenance activities within defence resource management system (DRMIS)	T	001
PT0290	Operate power generation system remotely	T	002
PT0280	Operate power generation system locally	T	002
PT0140	Operate electrical breakers	T	004
QT0390	Operate main diesel engine (MDE) remotely	T	002
QT0350	Operate main diesel engine (MDE) locally	T	002
QT0360	Operate main diesel engine (MDE) manually	T	002
QT0320	Operate diesel engine pre-heat system remotely	T	002
QT0310	Operate diesel engine pre-heat system locally	T	002
QT0320	Operate diesel engine pre-heat system remotely	T	002
YT0160	Operate shaft line mechanical emergency packing locally (sub)	T	002
YT0170	Operate shaft line mechanical seal locally (sub)	T	002
PT0300	Operate primary switch gear mechanical links locally (Subs)	T	004
PT0230	Operate motor generator locally	T	004

Number	Task	T/NT	PO
PT0240	Operate motor generator remotely	T	004
PT0120	Operate 440 VAC distribution system	T	004
PT0110	Monitor operation of 24 volt distribution system	T	004

Dates: February 6, 2017 – June 30, 2017

Location: Naval Training Development Center Atlantic

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Matte	KML	SLt	00211	TDC(A)
Wood	GK	PO1	00367.03	NFS(A)
Linzmeyer	KM	PO1	00367.03	NFS(A)
Cyr	CJ	PO2	00125.01	NFS(A)

Working Group Objective:

1. Create a QSP combining the SCC/MCC Watchkeeper course, the Victoria Class Mechanical Maintenance Course, and the Victoria Class Electrical Maintenance Course to be taught as one modular course.

2. Shorten the course duration.

3. Make all aspect of the course performance-based.

4. Modularize the course to ensure efficiency.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
	Supervise SCC2/MCC2 Watchkeepers	This task is performed based on rank, not qualification.
	Operate domestic fresh water system	This task is performed at the S2 level.
	Perform Watchkeeper Administrative Duties	Gaps were identified while scalaring. Therefore, new tasks were created in order to ensure all aspect of the performance is captured.
	Preparing preventive/corrective maintenance documentation	
	Conduct Engineering Rounds	
	Conducting on-watch rounds	
	Perform Diving Checks	
	Operate Diesel Generator	
	Preparing The Diesel-Electric Systems For Starting/Stopping	
	Starting/Stopping And Monitoring Correct Local And Remote Operation Of Diesel-Electric Systems	
	Operate Main Battery	
	Operate Main Motor	
	Operate Shaftline	
	Operate Sea Water Cooling System	
	Operate Fresh Water Cooling System	

Discussions / Decisions:

5. It was decided to not combine the Victoria Class Mechanical Maintenance Course and the Victoria class Electrical Maintenance Course due to the changes that are occurring within the new Mar Tech trade.

Lessons Learned:

6.

Guidance to Development and/or Instructional Staff:

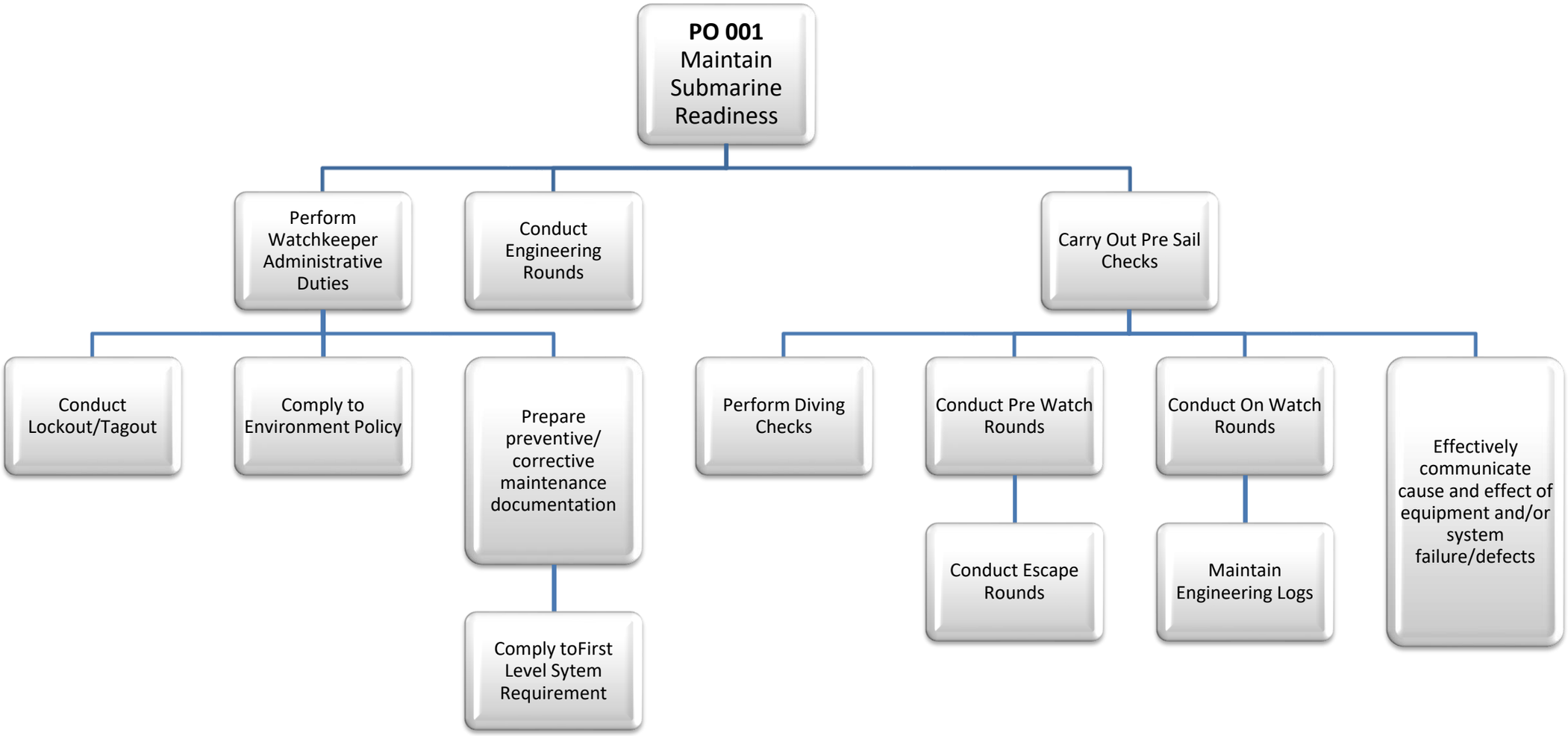
7. Pay attention to the requirement to divide the classes in half for training that requires the MCT and SCT trainers. The training can effectively train no more than 3 students at a time but they can be run concurrently.

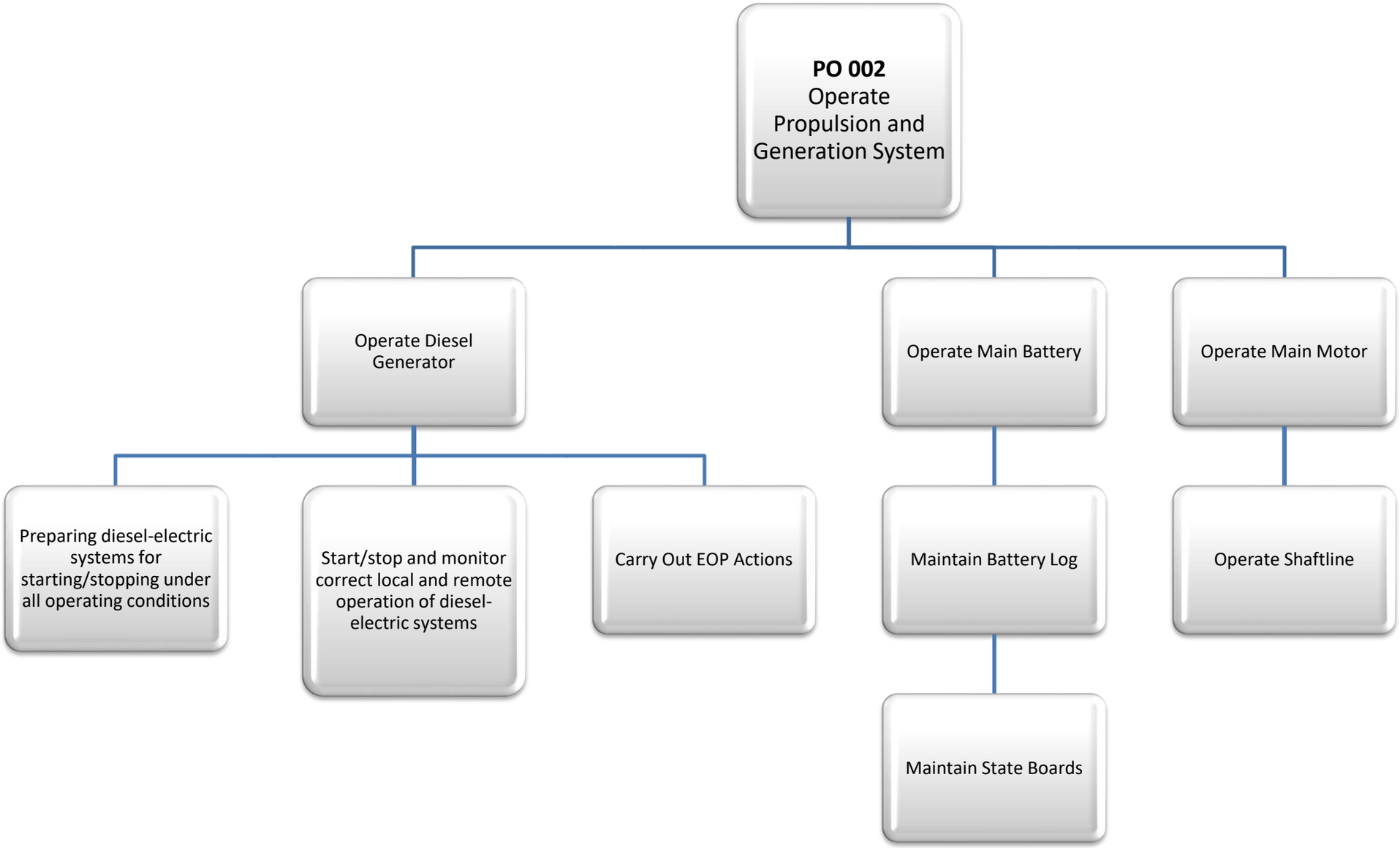
Follow Up Action:

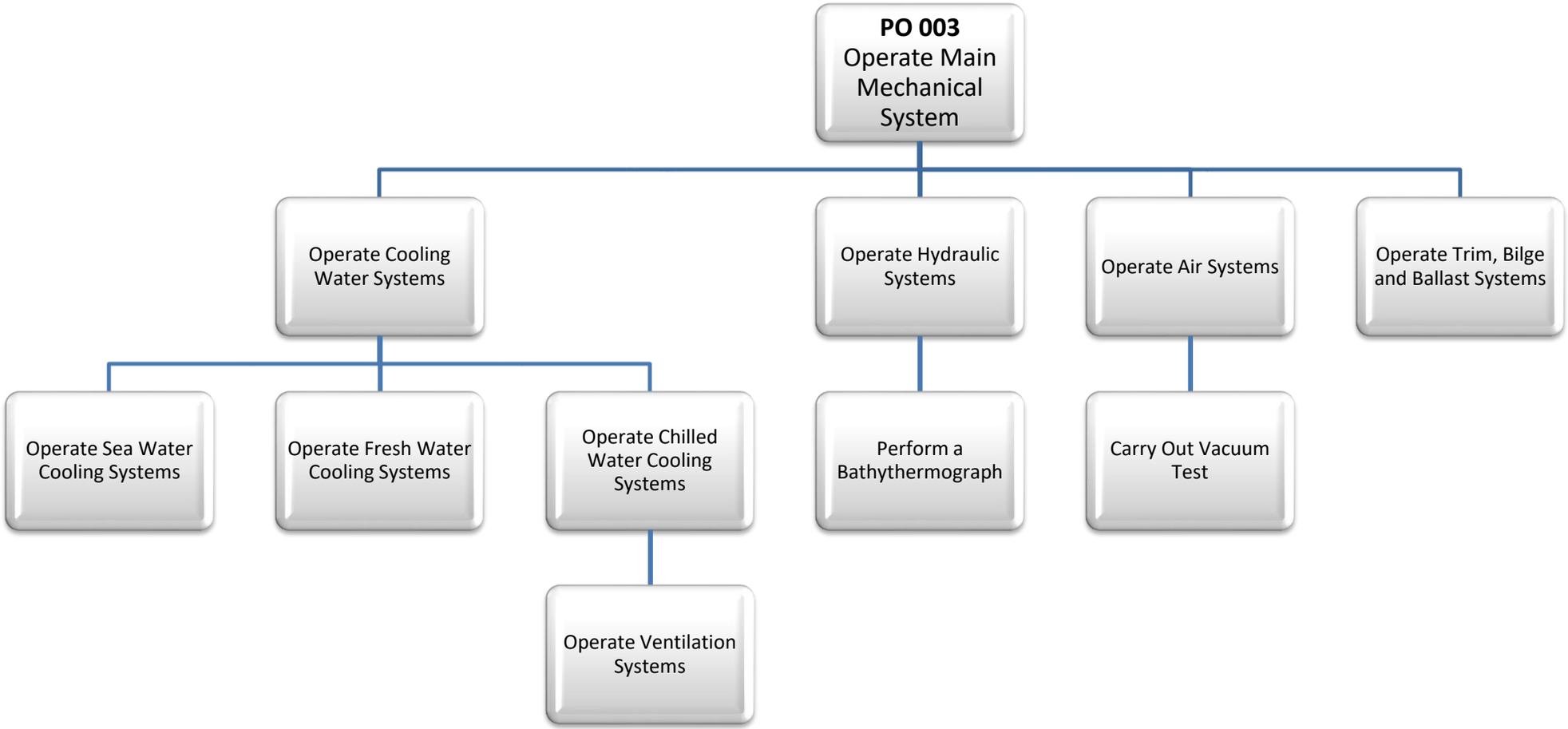
Item	Action	OPI	Expected Date of Completion
1			
2			
3			

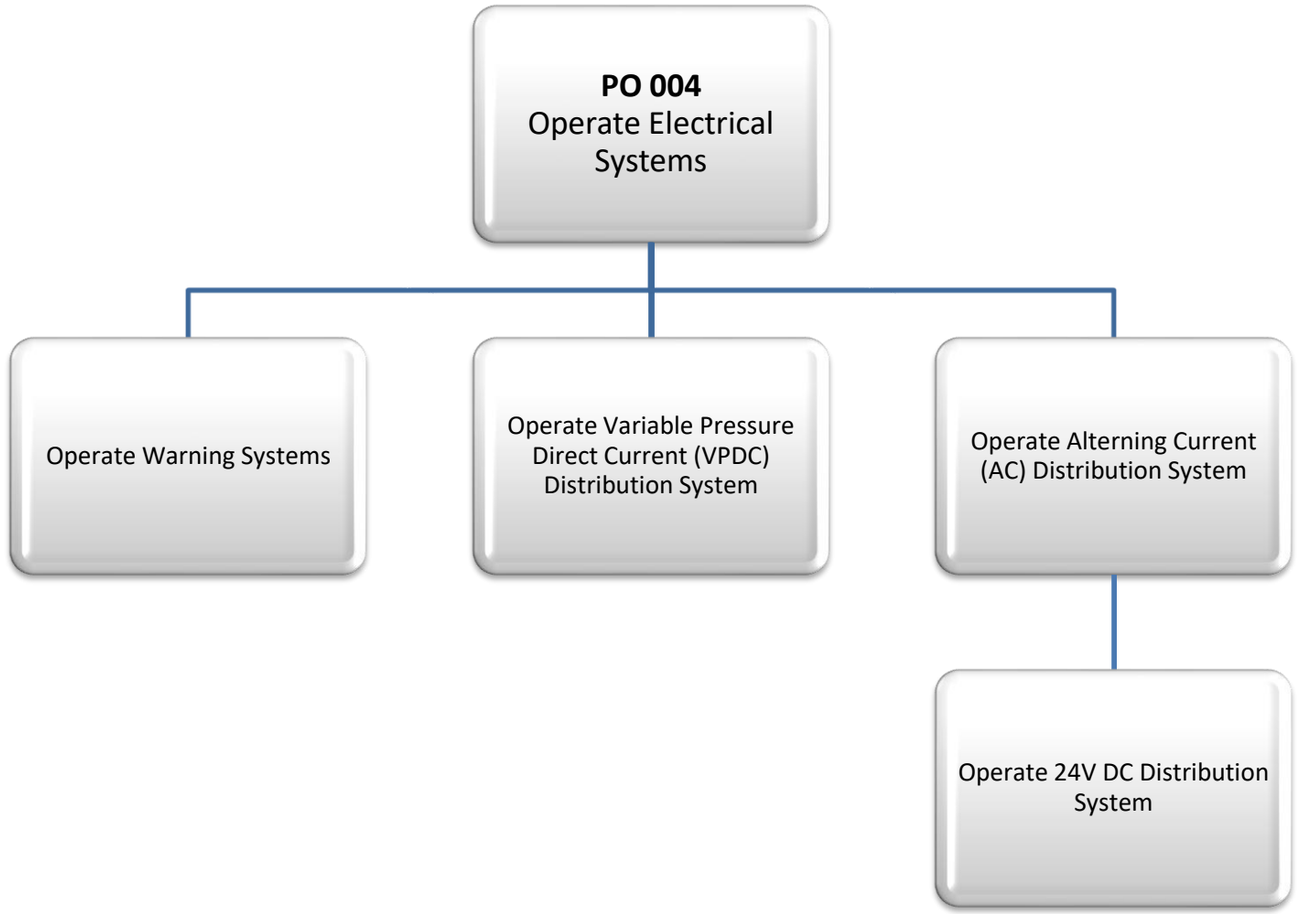
Other Considerations (if applicable):

28 Jan 2022 0900 – CPO2 Sutton TM, identified testing times may not be sufficient to evaluate condidates effectively. Notes added to review time and ensure that timings make sence. Added comments in teaching points to emphisizes communication or support of SME for opertaions of equipment.


PO 001
Maintain Submarine Readiness
Perform Watchkeeper Administrative Duties
Conduct Engineering Rounds
Carry Out Pre Sail Checks
Conduct Lockout/Tagout
Comply to Environment Policy
Prepare preventive/ corrective maintenance documentation
Comply toFirst Level Sytem Requirement
Perform Diving Checks
Conduct Pre Watch Rounds
Conduct Escape Rounds
Conduct On Watch Rounds
Maintain Engineering Logs
Effectively communicate cause and effect of equipment and/or system failure/defects


PO 002
Operate Propulsion and Generation System
Operate Diesel Generator
Operate Main Battery
Operate Main Motor
Preparing diesel-electric systems for starting/stopping under all operating conditions
Start/stop and monitor correct local and remote operation of diesel-electric systems
Carry Out EOP Actions
Maintain Battery Log
Operate Shaftline
Maintain State Boards


PO 003
Operate Main Mechanical System
Operate Cooling Water Systems
Operate Hydraulic Systems
Operate Air Systems
Operate Trim, Bilge and Ballast Systems
Operate Sea Water Cooling Systems
Operate Fresh Water Cooling Systems
Operate Chilled Water Cooling Systems
Perform a Bathythermograph
Carry Out Vacuum Test
Operate Ventilation Systems


PO 004
Operate Electrical Systems
Operate Warning Systems
Operate Variable Pressure Direct Current (VPDC) Distribution System
Operate Alterning Current (AC) Distribution System
Operate 24V DC Distribution System

PROTECTED B (when completed)

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

Certificate of Qualification

Name and Initials	Rank	SN

Date Training Started	Expected Date of Completion

NOTES

- The OJPR must be completed in accordance with the instructions below.
- Incorrect or incomplete OJPR will be returned for re-submission.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Naval Fleet Schools for guidance.

PROGESS REVIEW

4. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs),

OJPR Reviewed By:	Signature	Date

PROTECTED B (when completed)

On-Job Training Performance Record				
PC	**Title**	**Date**	**Signature**	**Comments**
SCC/MCC WATCHKEEPER				
001	Maintain Submarine Readiness		**(OJT Supervisor)**	
002	Operate Propulsion and Generator Systems		**(OJT Supervisor)**	
003	Operate Main Mechanical Systems		**(OJT Supervisor)**	
004	Operate Electrical Systems		**(OJT Supervisor)**	

5. Once all Performance Objectives have been completed, the trainee must pass a consolidated examination. The supervisor is to contact NFS(A) USQ Engineering Watchkeeper VIC senior instructor in order to get a standards exam. Upon successful completion of the exam both the MSEC and the EO will sign the part below.

SUCCESSFUL COMPLETION OF WRITTEN EXAM

_____
MSEC signature Date

_____
EO signature Date

6. The final portion of this OJPR is to successfully challenge an oral board. The Engineering Officer shall chair this board. At a minimum the other members will be MSEC, CET/Wrecker or designated representatives.

TRAINING COMPLETION CERTIFICATION

7. _____ has completed the requirements from the OJPR and shall be awarded the qualification for USQ Engineering Watchkeeper VIC (NQUAL), MITE Codes 122629 & 122630.

_____
Board Chairman Date

_____
MSEC signature Date

_____
CET/Wrecker signature Date

8. I certify that _____ has completed training to the standards stated in QSP USQ Engineering Watchkeeper VIC .

_____
Commanding Officer Date

PROTECTED B (when completed)

ON-JOB PERFORMANCE RECORD SHEETS

**** Note 1:** It is strongly recommended that these PO checks be completed at sea in order to satisfy the candidate's competency to the MSEC and EO. Although it is strongly recommended that these PO checks be completed at sea the EO/MSEC, with the CO's endorsement, will have the flexibility to check the candidate with a Submarine alongside or in the SCT/MCT.

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
PO 001 Perform Duties and Responsibilities of SCC/MCC Watchkeeper				
EO 001.01 Perform Watchkeeper Administrative Duties				
001.01.001	Conduct a Mechanical Lockout Tagout	(NQUAL MECH)		
001.01.002	Conduct an Electrical Lockout Tagout	(NQUAL ELEC)		
001.01.003	Conduct QA by Filling Out Appropriate OQE	(NQUAL)		
001.01.004	Create a Defect Notification (ND)	(SEWK)		
001.01.005	Complete PM Documentation (N017)	(SEWK)		
001.01.006	Conduct Mast Control and Man Aloft procedures	(DWS)		
001.01.007	Conduct Waste Management IAW SEMS	(SEWK)		
EO 001.02 Conduct Engineering Rounds				
001.02.001	Conduct SOP line-ups.	(Wrecker/Horse)		
001.02.002	Conduct Propulsion Pre-sea checks.	(Elec 2I/C)		
001.02.003	Conduct OMC pre-sail checks.	(OMC Maintainer)		
001.02.004	Conduct Vacuum Test.	(Wrecker/Horse)		
001.02.005	Conduct Vents and Blows pre-sail checks.	(Wrecker/Horse)		
001.02.006	Participate in Open Up for Dive	(Wrecker)		

PROTECTED B (when completed)

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
001.02.007	Participate in Long Post diving checks at MCC.	(Horse)		
001.02.008	Conduct Pre-watch engineering rounds.	(MCC1/SCC1)		
001.02.009	Conduct Escape rounds.	(MCC1/SCC1)		
001.02.010	Conduct On-watch engineering rounds.	(MCC1/SCC1)		
001.02.011	Participate in VCEOP's as SCC1.	(MSEC)		
001.02.012	Participate in VCEOP's as MCC1.	(MSEC)		
PO 002 Operate Propulsion and Generation Systems				
EO 002.01 Operate Diesel Generator				
002.01.001	Participate in Fueling (MCC)	(Fueling IC)		
002.01.002	Embark POLs (Lube oil, Hydraulic Oil)	(BLPO)		
002.01.003	Conduct snorting operations as SCC1.	(SCC1)		
002.01.004	Conduct engine prestart checks as MCC2.	(MCC1)		
002.01.005	Conduct load checks as MCC2.	(MCC1)		
002.01.006	Switch between fuel centrifuge and coalescers.	(MCC1)		
002.01.007	Demonstrate how to cross-connect air start reducers.	(MCC1)		
002.01.008	Operate surface muffler-exhaust mast change over valves.	(MCC1)		
002.01.009	Start Diesel Generator as MCC1 IAW VCSOP's.	(MCC1)		
002.01.010	Stop Diesel Generator as MCC1 IAW VCSOP's.	(MCC1)		
002.01.011	Conduct an engine start from the LCP.	(Horse)		

PROTECTED B (when completed)

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
EO 002.02 Operate the Main Battery				
002.02.001	State battery charging precautions and limitations.	(NQUAL ELEC)		
002.02.002	Describe types of charges.	(NQUAL ELEC)		
002.02.003	Monitor charging rates.	(MCC1)		
002.02.004	Maintain the battery log.	(MCC1)		
002.02.005	Operate Battery auxiliary equipment in both local and remote.	(NQUAL ELEC)		
EO 002.03 Operate Main Motor				
002.03.001	State power ratings of Main Motor.			
002.03.002	State parameters and limitations of Main Motor.			
002.03.004	Operate Main Motor in all modes.			
002.03.005	Operate Slow Speed MG.			
002.03.006	Monitor oil levels on the main motor bearings and thrust block.			
002.03.007	Operate shaft turning gear and shaft lock.			
002.03.008	Switch over to secondary mechanical shaft seal.			
002.03.009	Demonstrate Inflatable seal operation.			
PO 003 Operate Main Mechanical Systems				
EO 003.01 Operate Cooling Water Systems				
003.01.001	Operate Sea Water system in all modes.			
003.01.002	Operate Fresh Water Cooling system in all modes.			

PROTECTED B (when completed)

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
003.01.003	Operate Chilled Water System in all modes.			
003.01.004	Start and stop Chilled Water Plant.			
EO 003.02 Operate Hydraulic Systems				
003.02.001	Operate hydraulic plants in all modes.			
003.02.002	Carry out a Normal Repressurization.			
003.02.003	Extend and Retract the foreplanes using local hydels.			
003.02.004	Operate the Afterplanes and Rudder in all modes.			
003.02.005	Participate in cross-connecting hydraulic systems.			
003.02.006	Launch a bathythermograph from after SSE.			
EO 003.03 Operate Air Systems				
003.03.001	Operate HP Air Compressors in all modes.			
003.03.002	Switch over the Air Dryer			
003.03.003	Operate Emergency Blows using Pilot Valve and Remote (QA valves)			
003.03.004	Dive the submarine as SCC1.			
003.03.004	Surface the submarine with normal blows as SCC1			
003.03.005	Surface the submarine with emergency blows as SCC1			
003.03.006	Surface the submarine with the LP Blower as SCC1			
EO 003.04 Operate Trim, Bilge and Ballast Systems				
003.04.001	Operate the trim system as SCC1.			

PROTECTED B (when completed)

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
003.04.002	Operate the HP Bilge and Ballast system as SCC1.			
003.04.003	State parameters and limitations of HP Bilge/Ballast system.			
003.04.004	Transfer bilge contents ashore using LP Bilge Pump.			
003.04.005	Describe precautions when discharging oily-water to sea.			
PO 004 Operate Electrical Systems				
EO 004.01 Operate Warning Systems				
004.01.001	Acknowledge alarms on FDS.			
004.01.002	Reset alarms on FDS.			
004.01.003	Inhibit alarms on FDS.			
004.01.004	Monitor CSS parameters from an LSU.			
004.01.005	Create an Op Assist page on CSS.			
004.01.006	Inhibit and enable channels on CSS.			
EO 004.02 Operate Variable Pressure Direct Current (DC) System				
004.02.001	Operate battery switchboard locally and remotely.			
004.02.002	Operate DC Load Centres locally and remotely.			
004.02.003	Operate DG Switchboards locally and remotely.			
004.02.004	State all breaker ratings and trips.			
004.02.005	Transfer from Main Batteries to DC Shore Supply.			
004.02.006	Operate 24V TRU and distribution panel.			
EO 004.03 Operate Alternating Current (AC) System				

PROTECTED B (when completed)

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
004.03.001	Operate the 140KW MG in local and Remote.			
004.03.002	Operate the 140KW in hand control.			
004.03.003	Transfer power between Shore and MG supply.			
004.03.004	Operate AC Motorized breaker in hand.			
004.03.005	Describe procedure for inter-connecting Load Centres.			

Acronym or Abbreviation	Term or Definition
AAM	AUDIBLE ALARM MODULE
AC	ALTERNATING CURRENT
ACU	AUTOMATIC CONTROL UNIT
ATU	AIR TREATMENT UNIT
BSQ	BASIC SUBMARINE QUALIFICATION
CCA	CURRICULUM CONTROL AUTHORITY
DC	DIRECT CURRENT
NPTG	NAVAL PERSONNEL TRAINING GROUP
EFRU	EXCITATION FIELD RELAY UNIT
EMS	ENVIRONMENTAL
EOP	EMERGENCY OPERATING PROCEDURE
ETECH	ELECTRICAL TECHNICIAN
EXT	EXTINGUISHING MODULE
FDM	FIRE DETECTION MODULE
FW	FRESH WATER
FWD	FORWARD
GASH	GARBAGE
GIV	GROUP ISOLATION VALVE
HP	HIGH PRESSURE
HPB	HIGH PRESSURE BLOW
HZ	HERTZ
IPB	ILLUMINATED PUSH BUTTON
KW	KILOWATTS
LCD	LIQUID CRYSTAL DISPLAY
LCP	LOCAL CONTROL PANEL
LSU	LOCAL SCANNER UNIT
MARCOMHQ	MARITIME COMMAND HEAD QUARTERS
MARENG	MARITIME ENGINEER
MCC	MACHINERY CONTROL CONSOLE
MCT	MACHINERY CONTROL TRAINER
MG	MOTOR GENERATOR
MINERVA	FIRE DETECTION SYSTEM
MITE	MILITARY INDIVIDUAL TRAINING AND EDUCATION
MM	MAIN MOTOR
MPST	MAIN PROPULSION SWITCHBOARD TRAINER
MSEO	MARINE SYSTEMS ENGINEERING OFFICER
MTPU	MOTOR THERMAL PROTECTION UNIT
OJPR	ON JOB PERFORMANCE RECORD
OSLU	OVER SPEED LIMIT UNIT
PCL	AIR CONNECTOR
PD	PERISCOPE DEPTH
PGM	PROGRAMMING MODULE
QA	QUICK ACTING
QSP	QUALIFICATION STANDARD AND PLAN
RSU	REMOTE SENDING UNIT
SCC	SHIP’S CONTROL CONSOLE
SCT	SHIP’S CONTROL TRAINER
SSE	SUBMERGED SIGNAL EJECTOR

SSMG	SLOW SPEED MOTOR GENERATOR
SSWC	SHIPS SALT WATER CONSOLE
SW/FW PUMPS	SALT WATER/FRESH WATER PUMPS
SW/FW SYSTEM	SALT WATER/FRESH WATER SYSTEM
TRB	TRAINING REVIEW BOARD
TRU	TRANSFORMER RECTIFIER UNIT
V	VOLT
VCEOP	VICTORIA CLASS EMERGENCY OPERATING PROCEDURE
VCSOP	VICTORIA CLASS STANDARD OPERATING PROCEDURE
VDC	VOLT DIRECT CURRENT
VDU	VISUAL DISPLAY UNIT
VEX	VENTILATION MAST /SYSTEM
VPDC	VARIABLE PRESSURE DIRECT CURRENT

Module 1

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
EO 001.01	Perform Watchkeeper Administrative Duties	6	3		22
EO 001.02	Conduct Engineering Rounds	7	6		
EO 002.01	Operate the Diesel Generator	66	6		156
EO 002.02	Operate the Main Battery	29	3		
EO 002.03	Operate the Main Motor	42	3	7	
EO 003.01	Operate the Cooling Water Systems	5			65
EO 003.02	Operate the Hydraulic Systems	5			
EO 003.03	Operate the Air Systems	11			
EO 003.04	Operate Trim, Bilge and Ballast Systems	38	6		
EO 004.01	Operate Warning Systems	6			45
EO 004.02	Operate the Variable Pressure Direct Current (VPDC)	7	3		
EO 004.03	Operate the Alternating Current (AC) Systems	19	3	7	
	Total	**241**	**33**	**14**	**288**

* Include review and debrief time.

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**).

1. Include information about cost in whichever format is applicable (**Not to be deleted**).

Costing Annex_Template.xlsx

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Proj, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

1. This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details as to the nature and reasons for amendments are to be provided here.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	Perform Watchkeeper Administrative Duties	Scenario-based assessment	7/10 SATs and all critical items
EC 001.02	EO 001.02	Conduct Engineering Rounds	Performance-based assessment **(OJT)**	12/17 SATs and all critical items
OJT PC 001	PO 001	Maintain Submarine Readiness	Performance-based assessment **(OJT)**	18/18 Req's
EC 002.01	EO 002.01	Operate the Diesel Generator	Performance-based assessment	23/32 SATs and all critical items
EC 002.02	EO 002.02	Operate the Main Battery	Performance-based assessment	7/9 SATs and all critical items
EC 002.03	EO 002.03	Operate the Main Motor	Performance-based assessment	11/15 SATs and all critical items
PC 002	PO 002	Operate Propulsion and Generation Systems	Performance-based assessment	21/29 SATs and all critical items
EC 003.01	EO 003.01	Operate the Cooling Water Systems	Performance-based assessment **(OJT)**	6/9 SATs and all critical items
EC 003.02	EO 003.02	Operate the Hydraulic Systems	Simulated Performance-based assessment	5/7 SATs and all critical items
EC 003.03	EO 003.03	Operate the Air Systems	Simulated Performance-based assessment	5/7 SATs and all critical items
EC 003.04	EO 003.04	Operate Trim, Bilge and Ballast Systems	Performance-based assessment	6/8 SATs and all critical items
OJT PC 003	PO 003	Operate Main Mechanical Systems	Performance-based assessment	19/19 Req's
EC 004.01	EO 004.01	Operate Warning Systems	Performance-based assessment	6/9 SATs and all critical items
EC 004.02	EO 004.02	Operate the Variable Pressure Direct Current (VPDC)	Performance-based assessment	6/9 SATs and all critical items
EC 004.03	EO 004.03	Operate the Alternating Current (AC) Systems	Performance-based assessment	8/12 SATs and all critical items
PC 004	PO 004	Operate Electrical Systems	Performance-based assessment	17/21 SATs and all critical items

Information about development of Technology Enabled Learning (TEL) can be found in the embedded document:

Spirals.docx